



Guideline (Concept for recommended practice) for

Assessing the risk of undesired effects and interactions of large scale implementation of control and optimization algorithms in power systems.

Deliverable for TKI project GO-e (WP3 task4)

Version 1.1 (September 2024)

Version control

Version	Description	Date	remarks
0.1	Concept setup	September 2023	
0.2	Include STPA methodology	January 2024	
0.3	Include simulations	February 2024	
0.4	Draft for review	March 2024	
0.5	Added part on monolithic simulations	March 2024	
1.0	Final version guideline GO-e 0.1	April 2024	
1.1	Corrected minor spelling errors and added writers	September 2024	

Marcel Eijgelaar, Ravi Singh,
DNV Group Research and Development

DNV Headquarters, Veritasveien 1, P.O.Box 300, 1322 Høvik, Norway. Tel: +47 67 57 99 00. www.dnv.com

CONTENT

1	CONTEXT.....	4
1.1	The GO-e project	4
2	INTRODUCTION.....	6
2.1	The necessity and risks of automation within the power system	6
2.2	A guideline for identifying possible undesired effects	6
2.3	Governance and target audience for the guideline	7
2.4	Objective of guideline	8
2.5	Scope	8
2.6	Application	8
2.7	References	8
2.8	Key word descriptions (Definitions)	9
3	PRINCIPLES AND CONCEPT.....	11
3.1	Defining the system and its environment	11
3.2	Analysis methodology based on STAMP / STPA	11
3.3	Simulation and Modelling methodology	17
4	ALGORITHM ASSESSMENT.....	20
4.1	Objective of the algorithm	20
4.2	Input/output	20
4.3	Result	21
5	THREAT ASSESSMENT.....	22
5.1	Potential loss identification	22
5.2	Establish the control structure	23
5.3	Threat identification	24
5.4	Identifying loss scenarios	25
5.5	Analysis results	25
6	MODELLING GUIDELINES.....	26
6.1	Modelling objectives	26
6.2	Modelling loss scenarios	26
6.3	Modelling unknown interactions	26
6.4	Quantification of undesired effects	26
6.5	Quantification of the potential impact of undesired effects	27
7	REFERENCES.....	28
8	APPENDIX: SHORT EXAMPLE OF AN ANALYSIS.....	29
8.1	Algorithm assessment	29
8.2	Threat assessment	29
9	APPENDIX: MODELLING EXAMPLES.....	35
9.1	Bandwidth model	36
9.2	Statistical assessment of tariff – pricing interaction	39

Page 3 of 55

9.3	Home batteries	41
9.4	Increase-decrease gaming for redispatch markets	48
10	APPENDIX: LIST OF SYSTEM LOSSES IN THE POWER SYSTEM.....	51
10.1	Stakeholder analysis (ref to ENTSO-E)	51
11	APPENDIX: LIST OF KNOWN INTERACTIONS THAN CAN LEAD TO UNDESIRE EFFECTS	53
11.1	List of possible interactions	53
12	APPENDIX: DESCRIPTION OF THE RESULT IN THE GO-E PROJECT PLAN	54

1 CONTEXT

This document is a guideline for detecting and emulating undesired effects in complex power systems. It has been developed in the Dutch TKI project GO-E. This guideline is a draft, before it can become a DNV recommended practice (RP) it will have to go through both an internal and an external review process.

The document is about analysing and finding undesired effects of (new to be implemented) control and optimization algorithms, such as algorithms that are part of an energy management system. It does not offer solutions to counter these undesired effects, however it does offer insights that might help in countering these undesired effects. The scope of the GO-e project is on energy management and congestion, with a focus on energy flows. This initial version on this document has its focus therefore also on energy management. It is likely that the application of the final guideline will be more tailored to controls in the power quality and stability domain.

This guideline should be updated continuously. It should be expanded with more examples, especially in the power quality and stability domain (which was outside the scope of the GO-e project). After expanding this concept, it needs to be rewritten into a recommended practice (i.e. to be more formalized into rules and to be less reflective) and go through an extensive internal and external review process before being recognized as an official DNV RP.

1.1 The GO-e project

The first version of this document (v1.0) is established within the context of the GO-e project, which is supported by the Dutch government under the TKI/MOOI-support arrangement. The scope of the GO-e project is on disclosing flexibility of electricity demand and generation in the residential build environment, with a specific attention on avoiding grid congestion.

This first draft of the guideline for assessing the risk of undesired interferences of control and optimization algorithms, therefore also has its focus on the residential build environment and on possible undesired interactions related to congestion management and peak load. The document needs to be expanded with algorithms and interactions in the domain of system stability and power quality. As this document is about interactions of different algorithms, it is the ambition that eventually this guideline covers all electricity demand and generation, and transmission and distribution, as well as power quality, stability and (communication-)protocol issues.

1.1.1 GO-e deliverables

This document is the result of task 3.4 of the GO-e project. An overview of the results of this task is given below, a more comprehensive overview is given in appendix 12:

- Concept recommended practice (i.e. guideline, this document) for determining the possible negative side effects of a large-scale implementation of new optimization or control algorithms, including:
 - o Methodology for analysis
 - o Methodology and infrastructure for (hard and software in the loop) emulations/simulations
 - o List of categories of possible undesired effects (see appendix 11)
- Modelling methodology and infrastructure
 - o Some examples of the modelling of undesired side effects
- Gaming: increase decrease gaming (Technical University of Delft)
 - o Scientific papers with regard to increase decrease gaming

1.1.2 approach

The initial (exploratory) approach to tackle task 3.4 of the project was to implement actual algorithms that were to be developed in the project in a simulated environment and try to create the circumstances under which these undesired effects would emerge.

However, the descriptions of the flexibility dispatch mechanisms from task 3.1 that were to be used in the simulations turned out to be not sufficiently specific to be used in co-simulation-styled simulations. There are too many degrees of freedom in translating the dispatch mechanisms to actual algorithms implemented in control or optimization systems. There would have been a significant chance that possible undesired effects emerging in simulations would be the result of the decisions made during the translation of the mechanisms to code, instead of being inherent characteristics of the mechanisms themselves.

So, instead of working from the algorithms and simulations, the approach has been pivoted to possible undesired effects found in literature and experience, the analysis methods and replicating this in simulations. This resulted in the application of STPA to find and analyse potential undesired effect (see paragraph 3.2), and in a list of five categories of possible undesired effects that can be found in appendix 11. Some of these effects that are related to the scope of the GO-e project have been simulated in the co-simulation environment that was set-up. These can be found in Appendix 9.

2 INTRODUCTION

2.1 The necessity and risks of automation within the power system

This is a first guideline for assessing and analysing undesired effects of control and optimization algorithms in the power system. This document should be continuously updated and refined.

The power system needs to be—and eventually will be—increasingly more automated. Maintaining the required dispatchable generation capacity and grid capacity to be able to cover all possible realistic power flows no longer will be feasible, due to electrification of energy demand and the increase of variable renewable electricity generation.

The passive safety and reliability measures that currently are in place inevitably will be supplemented with more active measures that take into account the circumstances of the specific time and place and that make use of the potential synergies between the stakeholders in the system. Some of these existing passive measures, such as redundant generation capacity; N-1 redundancy for high and medium voltage networks; flat tariff systems might be replaced partly or completely, while others, such as circuit breakers, might become even more important as the system becomes more complex.

More active control, no matter how necessary, does come with new risks. Especially in a multi-stakeholder environment such as the power system that, in addition, is constantly evolving. Stakeholders might implement automation through algorithms that interfere with other algorithms or mechanisms, leading to undesired effects and even unsafe situations and injuries.

Having an insight in possible undesired side effects at an early stage is important. In an evolving and highly complex interconnected electricity system, undesired side effects might not emerge directly. They might only emerge after some time or after threshold has been met, and trying to reverse these effects might be very difficult and bring its own undesired effects.

2.2 A guideline for identifying possible undesired effects

The sooner undesired (side) effects of control and optimization algorithms are detected and identified, the easier it will be to avoid or reverse them. This guideline aims to be a first step in achieving the detection of potential undesired effects merging from interactions between different algorithms and mechanisms in the power system.

Active control and optimization is becoming more important, because of:

- (1) the power system is becoming increasingly more volatile because of the increase of variable renewables and electrification of demand. Swings in power demand and supply are becoming synchronized and much larger in magnitude. Dimensioning the network based on the largest possible (uncontrolled) load is becoming increasingly less feasible. Instead of controlling a small number of large scale generators tuned to demand, multiple smaller generators and large parts of the demand itself need to be controlled, each with their own specific boundary conditions.
- (2) Digitalisation allows for further optimization and take into account multiple (and local) boundary conditions and variables, such as the grid capacity (through congestion or capacity management schemes) and the cost to maintain the generation and demand balance, for example through specific commercial contracts and real time dynamic pricing schemes.

Examples of these developments are:

- Devices (loads) and generation governed by algorithms;
 - Dynamic (hourly or sub-hourly) electricity prices;
 - Power purchase agreements (PPAs) with specific conditions on capacity and/or flexibility;
 - Grid load managed by mechanisms, such as: congestion management; non-firm (non-guaranteed) capacity (with enforcement); different grid tariffs, cable and capacity pooling;
 - Imbalance management: (synthetic) inertia, FCR, aFRR, mFRR and BRPs
- trying to maintain a 15 minute average balance;
 - Other objectives of end-users, such as self-consumption;
 - Power quality and voltage control;
 - Sub-harmonic oscillations;
 - Regulation for safety, requirements for generators, etc.;
 - Forecasting algorithms;
 - Optimization algorithms, such as machine learning based algorithms, like reinforcement learning.

These, and many other possible kinds of algorithms will emerge in parts of the power system. They will be connected through communication channels, voltage and current measurements of the grid, pricing and/or market mechanisms etc. It is likely that they will affect each other, possibly in undesired ways. For example controllers reacting on each other creating instabilities; incentives synchronizing demand in a local grid and creating under or over-voltage or other power quality issues; implementation of artificial intelligence becoming unpredictable, e.g. reinforcement learning agents learning to exploit local grid congestion; gaming; or the monopolisation of (local) flexibility.

The ultimate goal is to prevent undesired effects caused by algorithms and mechanisms interacting with each other. However before it can be prevented, first the existence of an undesired effect needs to be identified (what effects might appear), qualified (under what kind of circumstances might they appear) and quantified (what are the chances and effects when these undesired effects they appear).

2.3 Governance and target audience for the guideline

How to prevent undesired interactions, and who is responsible for preventing these interactions is not well defined. While there are conditions and regulation with regard connections to the electricity network—for example requirements for generators that are implemented through the network codes—these are requirements for single connections. Distribution and transmission system operators are generally responsible for enforcing these conditions and regulation.

Undesired interactions by definition affect multiple connections and likely multiple stakeholders. It might be very difficult and even undesirable to address undesired interactions of algorithms by setting general conditions for individual algorithms. It is very likely that in practice such an approach would result in such strict conditions that the desired functionality of the algorithms will be greatly affected.

While it is unclear who is or who will be responsible for preventing undesired effect, it is likely that such an effect will affect multiple stakeholders in the electricity system, and all stakeholders have an interest in avoiding them. Mitigating actions will likely depend on the specific circumstances and the stakeholders involved, for example will the undesired effect only cause harm to the stakeholder responsible for the implementation of the algorithm causing it, or will the undesired effect foremost affect other stakeholders? And if this is the case, can the undesired effect reasonably be blamed to the specific algorithm, or is it more of an emergent effect?

A first step in preventing undesired effects caused by interacting algorithms is to foresee them and the circumstances under which they might appear.

2.4 Objective of guideline

The objective of this guideline is to provide guidance to assess the risk of undesirable interactions between a new control or optimization algorithm or scheme, and existing or anticipated control and optimization algorithms in the power system, when deployed at large scale.

2.5 Scope

Due to being part of the GO-e project, the initial focus of this guideline lies in application of control and optimization algorithms for small scale generation, storage and demand response in residential low voltage grids. However it is the ambition that eventually this guideline will be able to cover a wider range of algorithms and applications, such as for power quality and grid stability. While there might be more specific recommendations for these areas, the principles and key concepts will be applicable.

2.6 Application

The basic structure of an assessment of the risk that an algorithm will interfere with other parts of the power system is the following:

- Algorithm assessment:
 - o What is the algorithm's objective and its in and output
- Threat assessment
 - o Definition of relevant stakeholders with their objectives
 - o Identification of Hazards (defined as stakeholders not reaching their objectives)
 - o Definition of safety constraints
 - o Definition of the control structure guarding those constraints
 - o Assessment of control interactions
 - o Definition of loss scenarios caused by interactions

2.7 References

Relevant recommended practice that are associated with this guideline are:

- RP-A203 New technology Qualification
- DNVGL-RP-A204 Assurance of digital twins
- DNVGL-RP-0510 Data driven models
- DNV-RP-0513 Assurance of simulation models
- DNV-RP-0671 Assurance of AI enabled systems

2.8 Key word descriptions (Definitions)

Aggregator: Company aggregating the flexibility of end-users and representing them collectively by offering their flexibility to other stakeholders in the electricity system and electricity markets.

Algorithm: Any mechanism aimed to exercise control or influence within the power system. This can be dynamic, such as active control of a battery as a reaction to incentives and market mechanisms to passive elements, such as circuit breakers, fuses and fixed network tariffs, and anything in between.

BRP: Balance responsible party

BSP: Balance service provider

Control structure (STPA¹): A hierarchical control structure is a system model that is composed of feedback control loops. An effective control structure will enforce constraints on the behaviour of the overall system.

CSP: Congestion service provider

DSO: Distribution system operator

EMS: Energy management system (EMS)

End-user: Consumer of energy (might be residential, commercial or industrial and might also be producing, although generally not as its main activity).

Energy Supplier: Company selling energy (electricity and gas) to end-users, and taking care of the required administration (sourcing, billing, balance responsibility).

ESCO: Energy service company, a container name for companies providing any service around energy, ranging from maintenance of boilers to offering energy management or data services.

FMI: Functional Mock-up Interface defines a standardized interface for exchanging dynamical simulation models between different tools in a standardized format. It allows models to be exported and imported, enabling collaborative model-based development and system validation.

FMU: Functional Mock-up Unit is a model using the FMI standard.

Hazard (STPA¹): A hazard is a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss.

Loss (STPA¹): A loss involves something of value to stakeholders. Losses may include a loss of human life or human injury, property damage, environmental pollution, loss of mission, loss of reputation, loss or leak of sensitive information, or any other loss that is unacceptable to the stakeholders.

Loss scenario (STPA¹): A loss scenario describes the causal factors that can lead to the unsafe control actions and to hazards.

System (STPA¹): A system is a set of components that act together as a whole to achieve some common goal, objective, or end. A system may contain subsystems and may also be part of a larger system.

¹ All STPA definitions are from [3].

System-level constraints (STPA): A system-level constraint specifies system conditions or behaviours that need to be satisfied to prevent hazards (and ultimately prevent losses)

TSO: Transmission system operator

UCA (STPA): Unsafe Control Action (UCA) is a control action that, in a particular context and worst-case environment, will lead to a hazard.

3 PRINCIPLES AND CONCEPT

3.1 Defining the system and its environment

An undesired effect is loosely defined as an effect resulting from the interaction of a new control and optimization algorithm with already existing control and optimization algorithms, and that can result in a possible, but unacceptable loss for one or more of the stakeholders.

Defining the system boundaries for the analysis is a tricky task. Ordinarily the system is defined as being the responsibility of the developer of the system, and the systems environment consists of all aspects that the developer cannot (directly) control or influence. However while assessing the risk of the emergence of undesired effects caused by interaction algorithms, the whole system should be part of the analysis, not only the algorithm that triggers the assessment.

With regard to undesired effect in the power system, the responsibility of the system is shared along many stakeholders. The power system does not end with the transmission system with governed by the (transmission) system operator managing the (short term) system balance. It also includes the power markets that operate on timescales of 15 minutes (intraday markets) to years (futures and forwards markets) with all involved stakeholders, such as (dispatchable and variable) generators, end users, balance responsible parties; service providers, such as aggregators and energy service providers and their clients, as well as distribution infrastructure with their management of capacity and power quality. All of these stakeholders have an interest in the power system and might a reason to exercise some form of control to some part of the system.

In principle this guideline considers the system being everything that one or multiple stakeholders can control. The environment is everything that lies outside the control of the stakeholders of the power system. An example of possible elements of the environment and of the system are:

Example of elements that are part of the system under analysis	Example environment (that cannot be directly controlled)
Controllable demand (heat pumps, electric vehicles)	Non-controlled demand
The electricity network	solar irradiation
Solar panel/system output	(Outside) temperature
Inverters and power electronics	Holidays
Energy management systems	Weather
(Automated) trade strategies	Electromagnetic disturbances
	Most human behaviour

When assessing the possibility of the emergence of undesired effects of a particular control algorithm, it might be practical to limit the scope of the system.

For example it might not be very practical include a detailed model of the futures power market when assessing the impact of a reactive power control of a solar inverter on the voltage management in a local network. (Still, there might be interactions, such as the control algorithm might limit the active power output of a PV installation and therefore affect the long term position on the futures market).

3.2 Analysis methodology based on STAMP / STPA

Because this guideline is about undesired effects that are the result of *interactions* of algorithms/components, each of which can function perfectly fine, most traditional analysis techniques are not well suited as a basis for analysis.

Traditional methodologies focus on chain-of-failures events, root causes, and failure of—one or multiple—components. Examples of these methodologies are failure mode and effects criticality assessment (FMECA) and hazard and operability analysis (HAZOP). The methodology chosen to help analyse undesired interactions is STPA (System Theoretic Process Analysis), based on STAMP (System Theoretic Accident Model and Processes), which in turn is rooted in systems theory. STPA is able to analyse and predict possible undesired interactions of system components that by themselves perform as intended.

Instead of looking and preventing (component) failures, STAMP considers safety as an emergent system property that can be controlled by keeping the system within its safety constraints. This includes, but is not limited to, component failures. This paragraph gives a short introduction to STAMP/STPA based on the literature found in [1], [2] and [3]. More detailed information on the background and especially practical guidelines for application can be found in these documents, especially [3].

According to systems theory, a system is safe if it remains within its safety boundaries, called constraints. Accidents are the result of crossing these safety constraints. The associated methodology, STPA, does analysing the safety of a system by taking a systematic top down approach, and has the following steps:

- 1) defining the purpose of the analysis,
- 2) modelling the control structure,
- 3) identifying unsafe control actions, and
- 4) identifying loss scenarios.

Defining the purpose of the analysis consists of identifying losses, hazards and constraints on system level. This results in system constraints that should be to be controlled to avoid hazardous situations that could lead to possible losses under specific circumstances. The identified constraints on system level are the basis of the STAMP/STPA methodology. The rest of the STPA methodology consists of a systematic methodology to identify scenarios that can violate these constraints.

STPA seems to have a good fit to be used as a tool to analyse undesired effects that are the result of interactions of parts of the power system. However, a word of warning is in order. STPA is developed as a design tool to make sure that a designed system will be controlled in such a way that it will remain safe. The power system, including generation and demand, is not designed and managed by one authority, but is a collective effort. To an extend it has iteratively evolved more than that it has been designed. While STPA does acknowledge that systems and their environment will change over time, the power system (and its stakeholders) is a system that is constantly changing and evolving, especially considering the energy transition.

3.2.1 Identifying the purpose of the analysis and determine system constraints

A loss is something that is to be avoided and involves something of value to one or more relevant stakeholders, for example loss of live, loss of reputation, damaged equipment, financial loss, etc. Losses include the goals of the relevant stakeholders, such as:

- power loss/outage (loss of function) for the grid operators and end-user;
- the mission loss of a specific control algorithm, for example “charge the vehicle up to 80% before 7 am tomorrow for the lowest price”;
- mission goals of other relevant ‘algorithms’ of other stakeholders, such as algorithms/mechanisms (or tariff structures) to:
 - o prevent grid congestion (from the grid operator),
 - o prevent imbalance (from the system operator),
 - o maximize self-consumption (from the end-user and/or through a service of an energy service company/EMS provider),

- possible variable energy prices from the energy supplier to incentivise their customers to use energy when it is cheap or clean to produce,
- possible algorithms from other connected end-users, etc.

A (system) hazard is defined as a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to loss. A hazard related to a loss can be specified as <System> & <Unsafe condition> & <link to losses>. For example H-1 (Hazard 1) = “The power connection” (system) is “overloaded” (unsafe condition) leading to “damage of equipment” (loss)

This leads to system level constraints (SC) in the form of <System> & <Condition to enforce> & <Link to Hazard>. For example SC-1 = the power supply must remain within its voltage and current boundaries [H-1]. Constraints can also define how the system should react to minimize losses in the form of: SC = if <hazard> occurs, then <what needs to be done to minimize a loss> & <link to hazard>. For example SC-2= if “the power supply is overloaded” occurs, then “the power supply should switch off to avoid damage” [H-1].

So **Losses** (undesired effects) leading to **Hazards** (system states that can lead to losses under worst case circumstances), leading to **Constraints** (requirements for the system that should be continuously controlled). These constraints are the basis of STAMP based methodologies, such as STPA for hazard analysis and CAST (Causal Analysis based on System Theory) for the analysis of actual accidents.

3.2.2 Modelling the control structure

As mentioned, STAMP based methodologies consider the prevention of losses (in this case undesired effects) as controlling the system to remain within the limits of its safety, reliability and security constraints. To get insight in the system dynamics, a functional control model will need to be built. This control model is a conceptual model to identify the functional relationships and interactions, such as control actions and feedback related to the safety constraints of the system. It is not a computer model and might for example include humans.

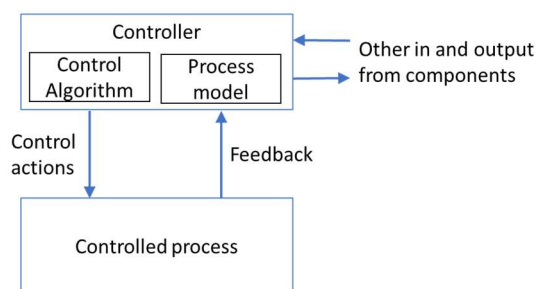


Figure 3-1, a basic control structure.

Based on the identified system constraints, basic subsections (controls) can be identified that should enforce these constraints. Each such control structure can be assigned a responsibility. A responsibility is basically a refinement of the system constraints. Together these should enforce the safety constraints on the system. Initially this can be quite abstract and high over. It can be refined later by zooming in on the responsibilities.

3.2.3 Identifying unsafe control actions

An unsafe control action (UCA) is a control action that, *in particular context and worst-case environment*, will lead to a hazard. There are 4 ways a control action can be unsafe:

- 1) Not providing the control action leads to a hazard.
- 2) Providing the control action leads to a hazard.
- 3) Providing a potentially safe control action but too early, too late, or in the wrong order.
- 4) The control action lasts too long or is stopped too soon (for continuous control actions, not discrete ones).

The description of an UCA consists of 5 parts and can be put in the following way (the order might change):

UCA-1: < Source > <Type> <Control Action> <Context> <Link to Hazard>.

For example: <EV charge controller> <provides> <charge command> <during peak demand in the grid> <link to hazard H1: overloading the power supply>.

After unsafe control actions (UCAs) have been identified controller constraints can be specified, with a link to the UCAs it is required to prevent. A controller constraint specifies the controller behaviours that is needed to prevent UCAs. For example: an EV charge controller must not provide a charge command during peak demand in the grid [UCA-1]. In principle each UCA can be inverted to define the constraints for the relevant controller/control structure.

Note, in addition to the 4 ways a control action can be unsafe, *if* an analysis is performed on a relative high level, such as for an energy management system, it is important to realize that there are lower level controls (or physical constraints) that might put limits on the control actions. So a control action might not be executed (or issued) because it gets overridden by a lower level control or physical limit that is not explicitly mentioned in the control structure (but would be if the control structure would be more detailed).

3.2.4 Identifying loss scenarios

After unsafe control actions and controller constraints have been established, loss scenarios can be identified. Two types of loss scenarios need to be considered: 1) unsafe control actions; and 2) looking control actions improperly or not executed:

- 1) Why would unsafe control actions occur?
This can occur because of:
 - a) Unsafe controller behaviour.
 - b) Inadequate feedback and information.
- 2) Why would control actions be improperly executed or not executed at all (leading to a hazard)?
Scenarios related to:
 - a) Scenarios involving the control path.
 - b) Scenarios related to the controlled process.

Figure 3-2 gives an schematic overview of this.

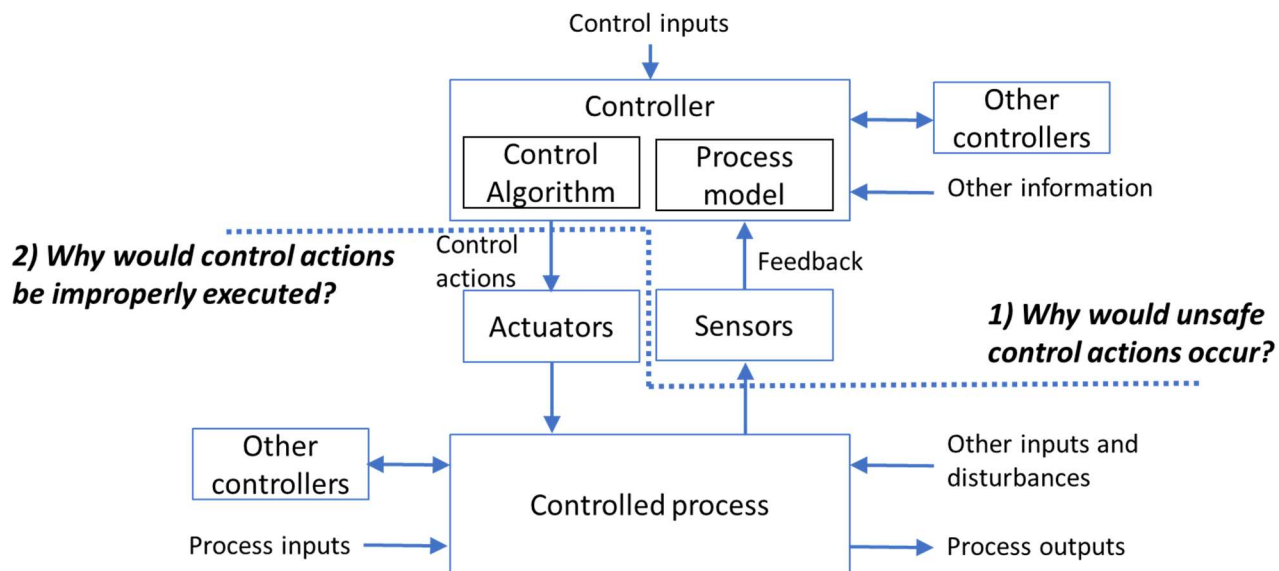


Figure 3-2, schematic overview of elements that must be considered to define loss scenarios. Each of the above right elements and arrows can lead to an unsafe control action. Each of the below-left elements and arrows can lead to a not or improperly executed control action.

Note, It is important to identify scenarios and not individual causal factors (such as 'sensor x fails to communicate feedback'), in order to maintain the view of how several factors interact with each other. Considering just single factors reduces the analysis to an FMEA analysis where only single components are considered.

3.2.5 STPA outputs and traceability

A properly executed STPA analysis will lead to a hierarchical 'structure' of losses, system level hazards and constraints, possible unsafe control actions and scenarios with and without unsafe control actions.

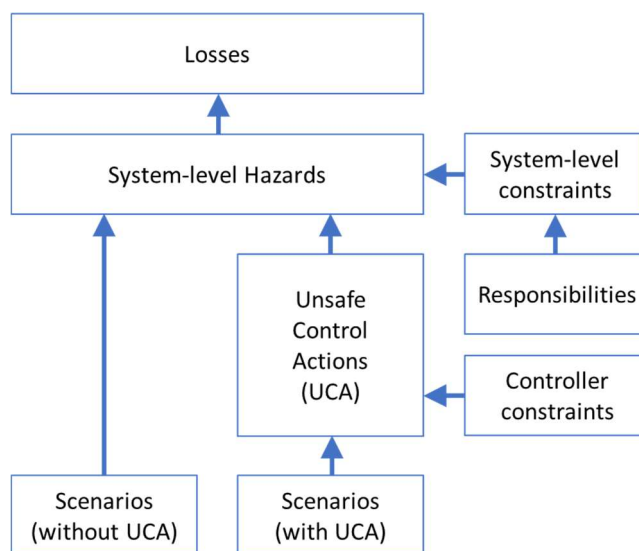


Figure 3-3, Overview of the outputs of an STPA analysis with their traceability

3.2.6 Applicability to the analysis of undesired effects of control and optimization in the power system

The STPA analysis can be very helpful in designing testcases and test plans to test control and optimization algorithms in the power system. For example to assess the impact of an (H)EMS system on the power system and if it interacts with other controls (in the wider sense, including incentive and tariff schemes). The initial focus of this document is on energy management and not on power management. However the electricity system has different layers of control which do affect each other. To keep the analysis comprehensible, lower level controls can be considered as boundary conditions or constraints instead of full control loops. Depending on the dept of the analysis, these can later be explored into more depth.

Section 5 will go into more detail in applying this methodology to the power system. Appendix 8 will give a short example.

3.3 Simulation and Modelling methodology

Purpose and scope of simulations

Within the scope of the assessment of the risk of undesired interaction between algorithms, modelling and simulations can have multiple goals, such as:

1. Further examination of predicted potential undesired interactions. For example possible interactions resulting from an STPA analysis, literature or through other means;
2. Examination of unknown interaction through large number of simulations. Depending on the complexity of the system this might result in a too large amount of possible situations and system states, this approach will not be able to exclude the existence of other unexpected undesired effects or undetected faults);
3. Quantification the circumstances of undesired effects, i.e. what is the risk that the undesired effect might appear and under what circumstances. For example to study when (part of) the power system becomes unstable, stability studies using simulations (or emulations based on Hardware in the Loop) can be performed, resulting in control-settings that minimize the risk of creating instabilities;
4. Quantifying the effect of an undesired effect if they appear. Some undesired effects can be tolerated until a certain point. Simulations can be used to estimate when that point might be reached.

Potential fail scenarios, whether or not detected through an STPA analysis, can be modelled and/or emulated using emulations and simulations on different levels of detail. Sometimes simulations are meant to assess and study specific emergent behaviour on a relatively high level, and sometimes the potential impact of specific algorithm and its interaction with existing algorithms in the power system needs to be assessed. The latter case requires the implementation of actual algorithms (e.g. as part of an energy management system, EMS) in an environment consisting of the other algorithms and controls with which it might interfere.

How to simulate and determine the level of detail in the simulation depends on the purpose. Demonstrating interacting mechanisms can be simulated on a relatively high level. Examples are evaluating the effect of a specific grid tariff mechanism on the optimized charging of electric vehicles, and the emergent behaviour of trading agents in a specific market. On the other end of the spectrum lies the testing a specific algorithm, embedded in power electronics like an inverter. This might require control hardware in the loop simulations in real time, where the specific physical device is coupled to a real time model of the rest of the (relevant) system.

So, while the modelling approach will depend on the purpose and scope of the algorithm, most applications will be somewhere in between the extremes stated above. For example, evaluating an energy management system (EMS) with embedded algorithms might require co-simulation in which the EMS is coupled to a simulated environment that consists of an ecosystem of other models, but that does not require to run in real time.

In the following, we distinguish monolithic and co-simulation approaches to simulation.

3.3.1 Monolithic simulation

Assessment of failure scenarios is often done within a single monolithic simulation environment that tightly integrates all algorithms. Such an approach is often used when assessing a specific failure scenario, for example to investigate the magnitude of the impact, the circumstances under which it occurs and possible remedies. In such cases, a special-purpose simulation may be constructed:

- Analytical simulation. In some scenarios, mathematical relations can be used to analyse possible failure scenarios without resorting to explicit simulation. For example, an analysis of incentives based on game theory can be used to identify structural problems in market designs.
- Special-purpose simulation. It is common to construct special-purpose simulations to analyse the behaviour of small systems (or small sub-systems). In the context of failure scenarios, such simulations are typically made to investigate a specific set of risks and model choices are made accordingly. For example, one may use a simple cost-optimising model for electric vehicle charging to investigate the potential synergies and conflicts between market incentives and grid tariffs.
- Agent-based simulation. A detailed simulation of behaviours of different agents (algorithms) can be constructed to investigate their interactions and possible emergent behaviour. Although this type of simulation may be constructed to investigate a particular interaction (previous item), it can also be used for more exploratory analysis.

3.3.2 Co-simulation

A drawback of monolithic simulation approaches is that it is necessary to implement all elements of the system under study within a single simulation environment. This can require substantial programming resources and may sometimes be impossible if implementation knowledge is inaccessible or restricted for IP reasons. Finally, failure modes may exist *because of* subtly different implementations that are more likely to show up in a heterogeneous environment. In such cases, co-simulation is a preferred alternative.

Existing algorithms, such as algorithms implemented in energy management systems, can be evaluated using co-simulation. Using the FMI standard (Functional Mock-up Interface), such an algorithm can interface with its simulated environment existing of other algorithms to demonstrate undesired interactions.

While limits to computing power might require simplifications in the modelling, this does not necessarily limit the usefulness of the simulations, provided the impact of simplifications in the modelling of the algorithms that form the environment of the assessed algorithm is considered carefully.

Many models of power electronic devices are already available in FMI format. New mock-up units (FMU, i.e. models using the FMI standard) can be developed that simulate specific behaviour in the power system (such as market prices, behaviour etc.).

It also allows for assessing 'black box models', i.e. undisclosed algorithms (or devices containing the algorithms) without the need for disclosure (although it should be recognized that with reverse engineering it remains possible to get insight in the algorithm, so the integrity and trustworthiness of the test organisation remains very important).

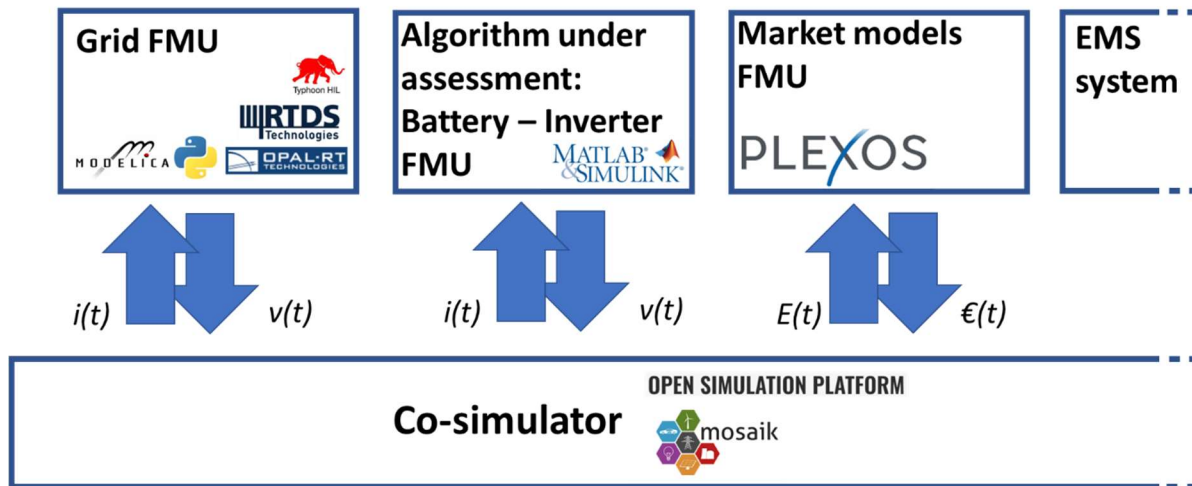


Figure 3-4, A possible way to use co-simulation to simulate a loss scenario caused by interaction of algorithms, represented by independent models (FMU's).

When independent algorithms are tested, such as the logic within an EMS system, Software In the Loop (SIL) is recommended, either real-time or through co-simulation using the FMI standard and including possible communication protocols. For co-simulation in simulated time, there are a number of tools that facilitate the coordination of the FMU models, such as Mosaik [4] and the Open Simulation Platform [5].

When results of the STPA analysis need to be simulated for further understanding, or when more abstracted (not coded) algorithms need to be assessed, such as generic optimization and/or tariff schemes, it is much clearer what needs to be modelled, and a choice for dedicated modelling tailored for the case will be appropriate.

Section 6 will go into more practical details about the purpose and level of detail of the simulations.

4 ALGORITHM ASSESSMENT

The setup of this guideline is built around the assessment of an individual algorithm. This means that the starting point of an analysis is the potential implementation of a specific algorithm in the power system, combined with a desire to analyse its potential impact.

The assessment has two major steps:

- Assessment of the specific algorithm
- Assessment of the interactions with other algorithms or mechanisms in the power system

After the qualitative assessment, simulations can be used to demonstrate undesired effects or show that possible undesired effects emerging from the analysis do not emerge. Simulations can possibly also be used to quantify both the risk of emergence of an undesired effect and its impact.

The analysis of an algorithm consists of:

- Its intended purpose or objective
- Its input, possibly also before deployment (such as data used to train the algorithm, settings of parameters, etc.)
- Its output (scope of control, influence and information)
- Quantities in the power system that might be affected (such as voltages, currents, harmonics, prices).

4.1 Objective of the algorithm

In principle each algorithm will have an objective. There might be exceptions where the objective is not very obvious, e.g. because the algorithm is obsolete, or the 'algorithm' or mechanism has evolved, such as some behaviour. However, for the algorithms being assessed by this guideline this will usually not be the case.

4.2 Input/output

Algorithms will have input and output. Mostly this input will be used during operation. However it might be that this input was used during the design of the algorithm. For example parameters, training data etc.

For example, a grid tariff system for end-users might be designed in such a way that it 'passively' influences behaviour, such as day/night tariffs that start at a fixed time and has a fixed fee per kWh, or a predetermined designed band width tariff with a progressively higher tariff per actually used kW. It might be that predetermined parameters are adjusted. For example, when the tariff increase for the higher bandwidth(s) is overruled by extremely high electricity market prices.

Algorithms will be connected by in and outputs. The control of a residential hybrid heat pump/gas boiler might look at the heat demand by looking at the difference between desired and actual room temperatures (and possibly also the outside temperature); electricity and gas prices and possibly requests/incentives from the grid operator, which can be fixed (like the forementioned band width tariff, requiring measurements from the total house hold electricity power demand), or can also be dynamic, like an situational incentive from the grid operator to reduce power (possibly even though an capacity buy back auction).

The output often will be a control signal or a signal to influence the system. As mentioned, it can also be a passive incentive, such as the aforementioned grid band width tariff, which is not influenced by changing circumstances in the grid (except in extreme circumstances).

The output of an algorithm will need to accomplish an objective. Not reaching this objective can be considered as a loss, (although it might be a minor loss).

4.2.1 Previous inputs and assumptions (e.g. data used for training)

Special care should be taken when considering non-feedback input, such as parameters and training data for machine learning algorithms.

Circumstances might change, so parameters might become no longer representative of the algorithm's environment. The process to updating parameters (or training data) is in principle a new feedback loop and should be considered as such.

For the algorithm assessment of machine learning algorithms DNV-RP-510-Data driven models can be applied.

4.2.2 Constraints of input and outputs

Are the input and output on the algorithm constraint, and if yes, how? Is this explicit, or implicit by other parts of the power system. What happens if the algorithm experiences extreme input (e.g. because of an error in its input). What happens if the algorithm gives an extreme output.

4.2.3 Quantities affected

Besides the direct output of an algorithm, other quantities might change as a result as well. An algorithm might change the load of a device, which in turn might change the voltage level in the grid.

Most of these quantities will be electrotechnical (voltages, currents, active and reactive power, frequency, sub and supra harmonics) and economical (costs, prices, risks) and/or related to information (data and information collection and exchange).

4.3 Result

This analysis should result in an overview of the algorithm that is being assessed and its direct environment, without diving into the algorithm itself.

5 THREAT ASSESSMENT

In the previous chapter the system was defined as the algorithm under assessment and its controlled influenced environment. In this chapter the system is enlarged to the whole electricity system to assess possible interferences with other parts of the electricity system, including other instances of the algorithm under assessment elsewhere in the electricity system. Depending on the level of analysis, large parts of the system can be abstracted though and explored later in a deeper analysis if required.

The electricity system is constantly evolving due to the energy transition, and a problem might rise arises when assessing the impact of for example a new control or optimization algorithm to influence electricity demand and generation. It might be perfectly safe now, but it might become problematic later due to developments in the system.

While not all potential problems can be found, some developments can be anticipated. For example when assessing algorithms to prevent grid overload, it is advisable to take into account some sort of mechanism from energy suppliers that is used to dispatch the available flexibility based on wholesale or balancing market prices. This can be direct control of flexible devices by the energy supplier or aggregator, or it can be a form of dynamic prices, which takes into account the cost of electricity on the wholesale markets.

Finally it should be noted that this approach is meant to find possible undesired effects of algorithms interacting with each other, after which measures can be taken to prevent them from happening. Because it is about interactions, it will not be possible to judge or blame an algorithm, except in rare circumstances.

5.1 Potential loss identification

5.1.1 Identify relevant stakeholders with their objectives

A list of possible stakeholders with their main objectives is shown in appendix 10.1.1. A selection can be made and supplemented with other identified relevant stakeholders not in the list (for example see 8.2.1.). Each relevant stakeholder has a set of objectives and related losses, i.e. not reaching the objective. These losses are input to the analysis, together with general losses, such as injuries or damages and possible 'loss of mission' losses of elements in the electricity system that do not come forward in the stakeholder analysis.

5.1.2 Hierarchy of objectives and prioritization of losses

Not reaching an objective of a stakeholder is in principle a loss. However, there is a priority in losses and some objectives will objectively have a lower priority than others. This is often represented already in the engineering of the power grid (i.e. the protection within a device or in a grid distribution station cut power in case of an overload, no matter what a higher energy management system desires). The priority of other objectives is more disputable (for example temporary overloading a grid station to allow for higher demand during oversupply of renewable electricity) and some will be very situational (for example the charging of EVs of two neighbours).

So unlike STPA analyses that are primarily about safety of (for example) a chemical installation, **a loss in this analysis might be (temporary) acceptable if this prevents a greater loss. However, such a situation should be flagged, because if such a situation becomes structural, then the**

situation is not 'stable'. Some sort of reaction will be triggered (that itself might be acceptable for the system or not).

For example: An energy management system of a battery on an industrial site, with an objective to arbitrage between the day-ahead and intraday electricity markets, might sporadically not be able to trade because the grid connection is used for the core industrial process. However, when this situation becomes structural, the battery might lose its function, or it might be beneficial to adjust the algorithms in the energy management system to include capacity management of the grid connection.

5.2 Establish the control structure

Each of the defined losses are related to hazards (i.e. the loss materializes). For each hazard the system has constraints that it should stay clear of, and which are governed by controls:

- Active controls: Often there are active controls in place to do this, such as voltage and frequency controls of (larger) generators or the Battery management system of a battery guarding against overheating of individual cells;
- Passive controls: Sometimes these controls are passive (such as circuit breakers and overvoltage protection of PV installations) and trigger when a certain constraint is reached;
- Statistics: Many parts of the electricity system are designed with a simultaneity factor smaller than one. It is (used to be) extremely unlikely that all connected devices will be active at exactly the same time, and this is factored in the in the dimensioning of the system.

Often though there will be a mix of different controls and there will be protection devices, such as circuit breakers to protect against injuries and damage of equipment. In principle a control structure should have a control or limitation for all identified hazards.

5.2.1 Levels of analysis

The purpose of the analysis determines the level the description of the controls that guard the systems constraints. Because the electricity system is large and consists of numerous controls, it is advisable to start relatively simple. This means that many controls will be abstracted to look as a seemingly physical constraint.

For example a battery might have a maximum storage volume. In a high over analysis (such as the example in appendix 8), this can be considered a 'physical' constraint (for example the maximum thermal energy content of a hot water tank with electric heater used to store heat).

However, in a more detailed level this constraint is enforced by controls on the hot water tank that make sure that the water in the tank does not cross a maximum temperature and pressure to avoid that hot water is vented (assuming a safety valve is in place to avoid that the tank ruptures).

5.3 Threat identification

5.3.1 Control action analysis and identify unsafe control actions

After the control structure has been established, it needs to be analysed and for each element the following questions need to be considered, according to the STPA methodology:

- 1) Not providing the control action leads to a hazard.
- 2) Providing the control action leads to a hazard.
(including excessive gain ('over-control'), e.g. caused by simultaneous behaviour of similar algorithms)
- 3) Providing a potentially safe control action but too early, too late, or in the wrong order.
- 4) The control action lasts too long or is stopped too soon (for continuous control actions, not discrete ones).

In case of a high level analysis (because of the scope of the electricity system), an additional point of attention is added with regard to the (seemingly or real) physical constraints on the high level analysis, and how these constraints might limit or prevent the possible control actions (or interfere with feedback-loops):

- 5) A (seemingly) physical constraint or lower level control preventing the proper execution of a control action (as mentioned in point 1 to 4).

5.3.2 Scan for known possible interferences

Besides the STPA analysis looking for possible unsafe control actions, it is also advisable to approach the problem from the other side and scan for possibilities that algorithms will interact with each other. Possible ways algorithms in the electricity system can interact with each other are:

- 1) *Triggers for simultaneous behaviour*, for example triggered or synchronized by an event or signal (such as a price signal). Special care should be taken when there is accumulation/storage and a chance for simultaneous release of energy.
- 2) *Second and higher order system control instabilities*
These are often already the study of power electronics.
- 3) *Undesired triggering or undesired failure of triggering of controls or protection* (e.g. anti-islanding, fault ride through).
- 4) *Harmonic interferences* (e.g. measurements based on sampling might result in wrong readings)
- 5) *Incentives counteracting each other without limitations/ increase-decrease gaming*. (e.g. incentives that bid for flexibility without any constraint)

Appendix 11 shows this list with examples. The control actions and feedback loops in the control scheme can be checked against these known interactions (if relevant).

5.4 Identifying loss scenarios

After possible unsafe control actions and possible interferences have been identified, the following questions need to be answered:

- Why would such unsafe control actions occur?
 - o Unsafe controller behaviour
 - o Inadequate feedback and information
- Why would control actions be improperly executed or not executed?
 - o Scenarios involving the control path
 - o Scenarios related to the controlled process

As mentioned before, lower level controls that are below the level of analysis likely will emerge as a constraint, which might limit control actions or feedback loops. These constraints, which might be time dependent, should be taken into account as well.

The answers to these questions will lead to scenarios (series of events, circumstances etc.) that lead to a loss, i.e. the combination of a hazard (system state) and the circumstances that this hazard materializes in a loss.

Note: In case of optimizations it is well possible that constraints that will limit the behaviour of an element in the future already will have an impact earlier. For example in appendix 8.2 an example is given of how an optimization algorithm of a home battery with a primary objective to store electricity to maximize self-consumption, might still discharge into the grid during an hour with a peak electricity price, in order to avoid having to feed the electricity back to the grid when prices are low and because the battery is full.

5.5 Analysis results

The SPTA analysis leads to a hierarchical structure of losses, system level hazards and constraints and unsafe control actions and scenarios with and without unsafe control actions.

The main result is an overview of loss scenarios of a new control or optimization algorithm in the electricity system interacting itself when implemented at a large scale and with other controls and/or optimization algorithms in undesired ways. It does not give immediate solutions to remedy this, however it does give insight in why these mechanics appear. Therefore this approach can help in identifying possible undesired interactions before they appear, and to design counter measures to make sure they don't.

Automated control and optimization in the electricity system is essential for the energy transition to succeed, and this approach is not intended to prevent this. Instead the intention is to find possible undesired effects before they do harm and when solutions to prevent these undesired effects to happen can be found, while maintaining the desired effects of the algorithms.

If required these loss scenarios can be further explored, investigated and (though limited) quantified using simulations and emulations.

6 MODELLING GUIDELINES

6.1 Modelling objectives

The objectives of the modelling should be clear. The objective will determine the methods and level of detail required in the modelling. On a high level modelling can have one or more of the following four objectives:

- Modelling a known or a potential loss scenario
- Modelling unknown interactions
- Quantification the circumstances of undesired interactions (e.g. to determine protection settings)
- Quantifying the impact of known undesired interactions

6.2 Modelling loss scenarios

Modelling a loss scenario identified in the STPA analysis gives more insight into the loss scenario and possible solutions that prevent the scenario to materialize. Sometimes an identified potential effect emerges only under specific circumstances and with specific relative sizes of demand and generation. (for example the failure of anti-islanding protection of residential solar systems, because above a certain number of installations, the systems feed each other, and fail to detect an grid outage).

If the dynamics of the interaction is to be understood, the modelling often will benefit of simplifications (i.e. to leave out the elements that do not contribute to the interaction). The more quantitative analysis the more accurate the modelling should be (see also 6.4 and 6.5.)

6.3 Modelling unknown interactions

Especially in the electrotechnical stability domain, modelling can be used to determine if undesired effects, such as harmonic interactions, will appear. This requires a well described systems and control algorithms in inverters and other power electronics. While some stability effect will be unexpected, most will be known that they might appear.

Because the level of detail of the modelling will determine whether an effect will or will not appear. In very detailed modelling, the number of possible states might be extremely high. This means that it is unlikely that all possible and unexpected interaction will be found.

6.4 Quantification of undesired effects

Quantification of when an undesired effect will emerge and under what circumstances this can be prevented. This involves advanced, but well established and electro-engineering practices, such as for stability calculations, advance protections settings etc.

An example of these practices are the calculations and modelling for grid forming functionality of offshore wind farms, that are used to provide stability services to the grid through the algorithms in the inverters (inertia, FCR and black start functionalities). In the absence of fuel powered generation with rotating mass, such as traditional powerplants, these services need to be provided by other assets (i.e. predominately of inverters of renewable generation and inverters of storage systems)

6.5 Quantification of the potential impact of undesired effects

Quantifying the impact of an undesired effect is challenging because it will depend on numerous variables. what can be determined to a certain extent is the sensitivity of the impact to these variables. For example the (relative) number of the algorithms that is being assessed that are implemented, the (relative) number and ratios of other algorithms, different loads and grid strength and stability, price fluctuations, tariff schemes etc.

7 REFERENCES

- [1] N. G. Leveson, „A New Accident Model for Engineering Safer Systems,” *Safety Science*, vol. 42, nr. 4, pp. 237-270, April 2004.
- [2] N. G. Leveson, „CAST Handbook: How to learn more from incidents and accidents,” 2019. [Online]. Available: <https://psas.scripts.mit.edu/home/materials/>.
- [3] N. G. Leveson en J. P. Thomson, „STPA Handbook,” 2018. [Online]. Available: <https://psas.scripts.mit.edu/home/materials/>.
- [4] OFFIS, „Mosaik,” 2023. [Online]. Available: <https://mosaik.offis.de/>.
- [5] Open Simulation Platform, „Open Simulation Platform,” 2023. [Online]. Available: <https://opensimulationplatform.com/>.
- [6] fmi-standard, „<https://fmi-standard.org/>,” [Online]. Available: <https://fmi-standard.org/>.
- [7] N. G. Leveson, *An introduction to system safety engineering*, Cambridge, Massachusetts: MIT Press, 2023.
- [8] ERI Grid, „<https://erigrid.eu/>,” May 2017. [Online]. Available: https://erigrid.eu/wp-content/uploads/2017/07/DL_D-NA5.1_ERIGrid_Smart_Grid_configuration_validation_description_method_2017-05-01.pdf.

8 APPENDIX: SHORT EXAMPLE OF AN ANALYSIS

In this appendix a (rather high over) algorithm of an (hypothetical) energy management system for a home battery is being assessed.

Residential batteries are sometimes being advertised as a solution to the voltage problems that are related of having too much installed PV panels on a low voltage power cable in a residential area. On a sunny day with little demand, the collective PV panels will cause a rise in voltage in the grid (to export the generated electricity) and will lead to the disconnection of panels because overvoltage protection is triggered.

8.1 Algorithm assessment

Algorithm to be assessed: Home battery optimized for self-consumption of PV (and how it relates to capacity management).

Objective:

- 1) First priority: For an end-user, use as much energy from the rooftop PV panels as possible for own use;
- 2) Second priority: With the remainder: optimize the electricity bill.

Input (assessed every minute):

- 1) PV generation [Wh/minutes]
- 2) Power exchange with the electricity network [Wh/minute]
- 3) Electricity retail price [Eur/kWh changed every hour, based on the day ahead market, known 1 day in advance]
- 4) Wheater forecast (known 1 one day in advance).
- 5) State of charge of the battery (optional)

Output (updated every minute):

- 1) (Dis-)charging behaviour of the battery [Wh/minute]

The algorithm will forecast the PV yield and electricity use of the coming day based on historical data and the weather forecast. It will use this to save as much electricity as possible for own use but will use the rest to optimize the electricity bill.

8.2 Threat assessment

8.2.1 Identify relevant stakeholders with their objectives/hazards and constraints

List with relevant stakeholders and their objectives with regard to energy management. This list does focus on energy management related issues and does not include objectives and constraints in in the power domain (such as safety related issues, short circuits, etc). These basically function as a safety net and the energy management controls should avoid triggering this safety net. A more elaborate focussed on safety and reliability might include this.

Some stakeholders that in some circumstances might be relevant have been left out, such as the provider of the Battery EMS, the Transmission System Operator (TSO) and regulator for simplicity.

Main stakeholder	Objectives	Hazards	Tools/mechanisms
End-user	1) Maximize self-consumption 2) Minimize cost for electricity (tax, grid fee and energy cost)	Limited Self-consumption High cost for electricity	Behaviour, using battery dispatch algorithm of service provider
Other end-users (neighbours)	1) Minimize cost for electricity (tax, grid fee and energy cost)	High cost for electricity	
Distribution grid operator	1) Operate the grid within its safe and reliable boundaries (voltage and current constraints) 2) Minimize grid losses	Power outage (i.e. triggering protection) Component degradation Grid losses	None (statistics); Bandwidth tariff model, Capacity buy-back.
Energy supplier	1) market share (be attractive to customers) 2) Healthy margin (fixed monthly service fee) 3) Minimize merchant risk (wholesale prices with retail prices)	loss of reputation (and market share) Financial loss Exposed to high market risks	Electricity price scheme: Fixes monthly service fee + wholesale market price (hourly)

8.2.2 Establishing the control structure

The chosen system is a neighbourhood. The main processes that are considered in this system are power generation and demand at the end-user(s) and the elements that influence this:

Processes (for multiple households):

- Electricity generation with PV.
- Electricity demand by appliances and lighting
- Battery storage:
 - Charging
 - Discharging

As mentioned, lower level controls guarding safety are not included in the initial schematic control structure but are included as constraints. A more detailed analysis could include them.

Constraints by physics or lower level controls:

- Battery charge/discharge power [kW], limited by the battery management system.
- Battery volume capacity [between empty and full], limited by the battery management system.
- Household connection capacity, limited by a circuit breaker or fuse.
- Demand of individual appliances is limited by their capacity
- PV generation is limited by its capacity/capacity of the inverter.
- Grid load at distribution transformer, limited by protection gear/ circuit breakers
- Grid voltage at household, limited by PV inverter overvoltage protection (and possibly the battery inverter overvoltage protection).
- Possible (contractual, legal or technical) limitations on the grid and retail tariffs.

Figure 8-1 shows the basic energy management control structure for one household. Note, there are many controls that are not incorporated in this diagram, such as the power point tracker in the PV inverter, several protection control in the battery management system, the circuit breakers in the household connection and protection further in the grid, etc.

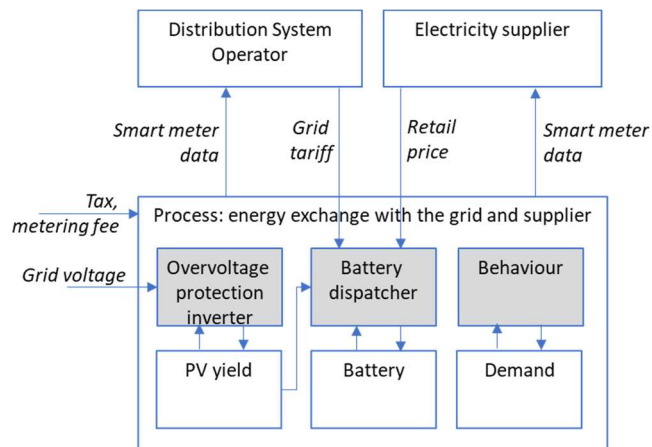


Figure 8-1, Schematic overview of the basic energy management control structure within the household.

Figure 8-2 shows the control structure of the energy flows one level higher than Figure 8-1, incorporating the neighbourhood. Different end-users might have contracts with different energy suppliers and/or aggregators and have different contract conditions and energy pricing, such as a fixed price per kWh, or a battery that is used by an aggregator for offering balancing services to the transmission system operator (not shown in the figure). For simplicity only fixed kWh contracts and dynamic wholesale market based retail prices are considered.

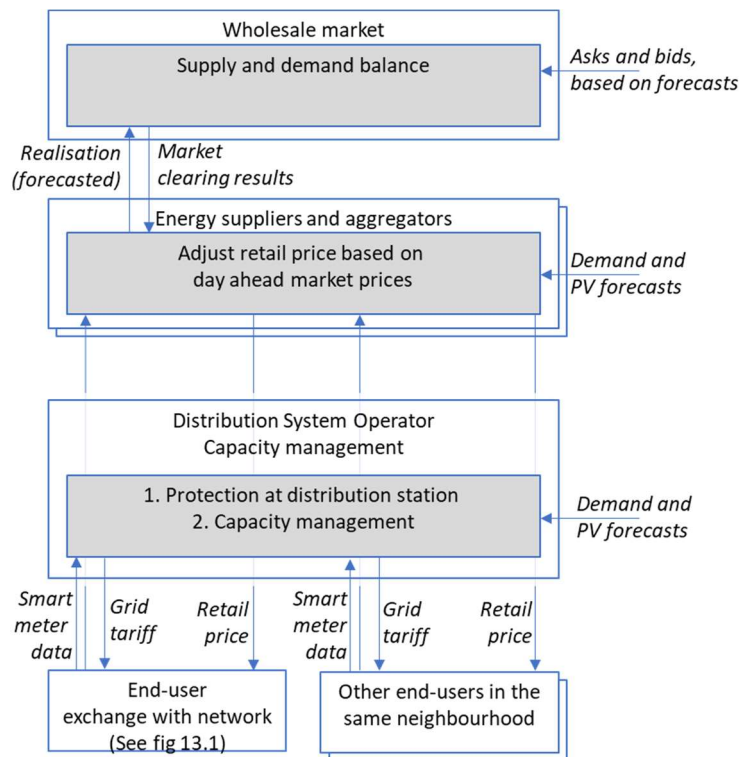


Figure 8-2 Schematic overview of the control structure of the electricity management of a residential neighbourhood. Note, there is in principle no control hierarchy between the energy suppliers and the Distribution system operator.

Control of energy on this level include the protection of the grid distribution station feeding the area, which protects the equipment against over damage, and the energy balance on the wholesale market that governs the wholesale prices, and—in this example though dynamic pricing contracts—also the retail contracts. Historically, the distribution grid system is designed based on historical data based on statistical load (usually a simultaneity of load of 20% to 40% was considered).

Historically energy contracts were based on a fixed kWh price, independent of the time of use. The demand of (a large group of) households is well predictable using historical data, and levels out very well.

8.2.3 Analysis of the control structure

This step consists of identifying unsafe control actions:

- 1) Not providing the control action leads to a hazard.
- 2) Providing the control action leads to a hazard.
- 3) Providing a potentially safe control action but too early, too late, or in the wrong order.
- 4) The control action lasts too long or is stopped too soon (for continuous control actions, not discrete ones).

As mentioned, this analysis is on the relative high energy management level and does not include most safety controls that are included on the power level, such as leakage protection or short circuit protection. However, triggering these protections will result in an outage, so the energy management

controls should steer clear of activating these protections. I.e. for this analysis triggering these protections are considered to be a hazard. For the analysis it is important to include where a control action cannot be executed because of the constraints provided by these lower level controls.

When looking at the control structure in Figure 8-1 and Figure 8-2, the following observations can be made:

- It is obvious that all stakeholders (the end-users with its PV installation and demand, the grid operator and the energy supplier) do control or influence the (dis-)charging behaviour of the battery. The end-user by setting the strategy of the battery on maximizing self-consumption, and by his PV generation and electricity demand; the energy supplier through dynamic pricing, and the grid optionally through a grid tariff (or other signal). Note, this is not necessarily an issue, provided the controls either have a hierarchy or if they are limited.
- The retail price of all end users (with a dynamic electricity price has the same origin, the wholesale market.
- A potentially unsafe control action or feedback leading to an overvoltage or overcurrent on individual household level will be contained by lower level controls (i.e. the PV overvoltage protection and the circuit breaker in the household). However other disturbances, on power quality level, such as voltage dips and harmonics might not. (This is outside the scope of this example analysis. To include this would require a more detailed control structure analysis).
- The control actions can be limited or overridden by lower level controls and protection, such as the battery management system indication that the battery is full or depleted, or that the charging capacity (kW) has been reached.

Looking at the control actions and their constraints, it becomes clear that if the battery limitations are reached and its operation becomes limited, it can no longer be operated purely based on self-consumption. E.g. if the battery is depleted, it can no longer supply the energy to the household, and the household needs to draw electricity from the grid. If the battery is full, it can no longer absorb the produced solar electricity and it needs to feed the remainder to the grid.

However, if self-consumption can no longer be further maximized, the algorithms in the battery energy management system will try to optimize the electricity bill. It might already start to discharge before the battery is full, if the electricity retail price is high and it knows that soon more PV power will need to be stored. If the case if the battery is expected to be depleted and more electricity is needed, the energy management might decide to charge at an appropriate time when prices are low.

In both situations the battery will behave based on external incentives from the Energy supplier (the dynamic electricity price) and—assuming this is also dynamic—the grid tariff. Zooming out that the neighbourhood level, it becomes clear that this applies to all households in the neighbourhood that have a dynamic retail price. The retail price is a trigger for simultaneous behaviour. However, the same is true for a dynamic (time dependent) grid tariff (unless they cancel each other out).

8.2.4 Identifying a loss scenario

So assuming that the PV installation is sized to cover average demand, and if the battery is sized to cover average daily demand, then on average the battery should be able to cover demand. However, in summer there is structural oversupply of solar power. While the battery is sufficiently large to cover the household demand, the structural surplus needs to be fed into the grid (or the PV generation needs to be curtailed). In winter, the same system will have a structural shortage of solar energy, and while the battery still can cover the household demand, it needs to charge from the grid to do so.

The retail electricity price will cause batteries in nearby households to have the same behaviour. Whether or not this simultaneous behaviour is problematic for the grid operator depends on the amount of electricity that is drawn of fed into the grid, so the number of households with such a system. It also

depends on the influence the grid operator can exercise and on possible other constraints, such as constraints on the (dis-)charge power.

A grid bandwidth tariff (e.g. the first 2 kW is relatively cheap, while drawing power or fee-n power above the 2 kW becomes much more expensive) might work in most instances, provided the penalty of using the higher capacity bandwidth is higher than the revenue of selling the electricity all at once. However, when there are even a few instances where this is not the case, the retail price will 'override' the grid bandwidth tariff, creating simultaneous behaviour of the batteries and possibly causing an outage by triggering the protection of the distribution station.

Note, this analysis did not include equipment failures. To properly analyse the effects of these failures this analysis should be performed on a more detailed level, including the protection and safety devices.

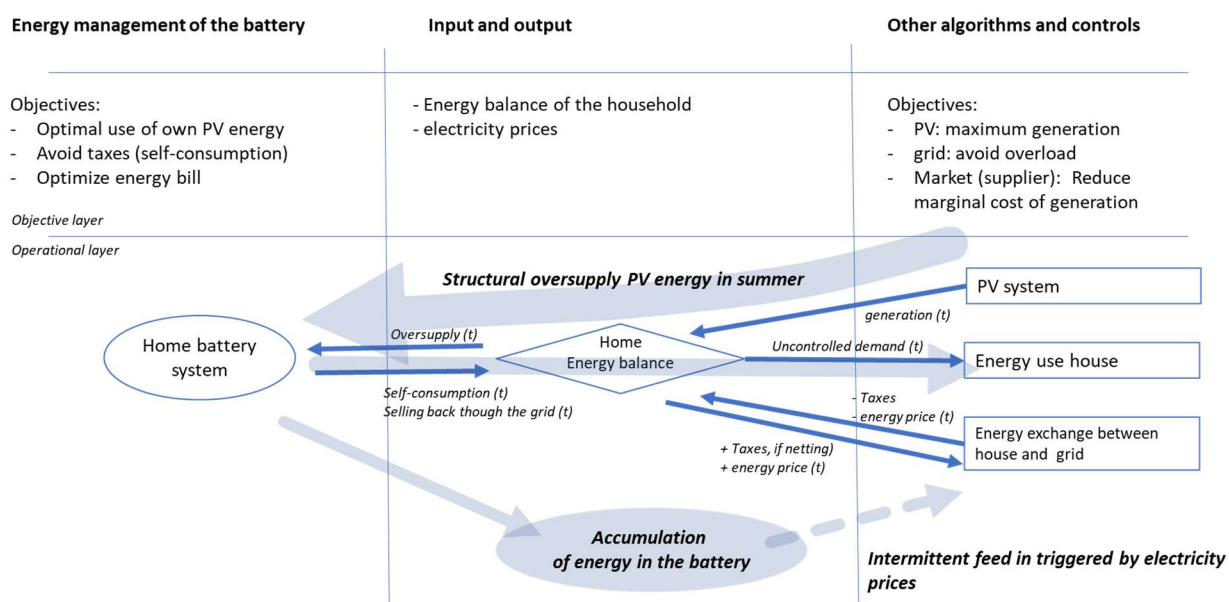


Figure 8-3, Overview of the loss scenario in summer. A structural oversupply of PV energy might be accumulated in home batteries of different households and intermittently, but simultaneously discharged into the grid when prices are high.

9 APPENDIX: MODELLING EXAMPLES

This appendix presents modelling examples to demonstrate two undesired effects: (1) a bandwidth grid tariff meant to discourage peak demand for EV charging getting overruled by market prices; and (2) residential batteries meant to increase self-consumption from rooftop solar panels, synchronizing the grid load for supplementary/surplus energy due to dynamic prices (see appendix 8). A co-simulation-based approach is utilized to simulate the electricity distribution grid and other components like household-loads, solar PV, EVs and household batteries. Mosaik platform which provides a flexible smart-grid scenario development and co-simulation framework has been utilized.

Pandapower (an open-source power system analysis tool with validated network components) is used to build the network and perform the power-flow calculations. A modified version of a suburban-type, low voltage Kerber network (which comes predefined in Pandapower) is used in the simulations. The distribution transformer supplying electricity to the grid is rated for 200 kVA power. The loads of the original network are removed and replaced with custom household loads. The layout of the network is shown in Figure 9-1.

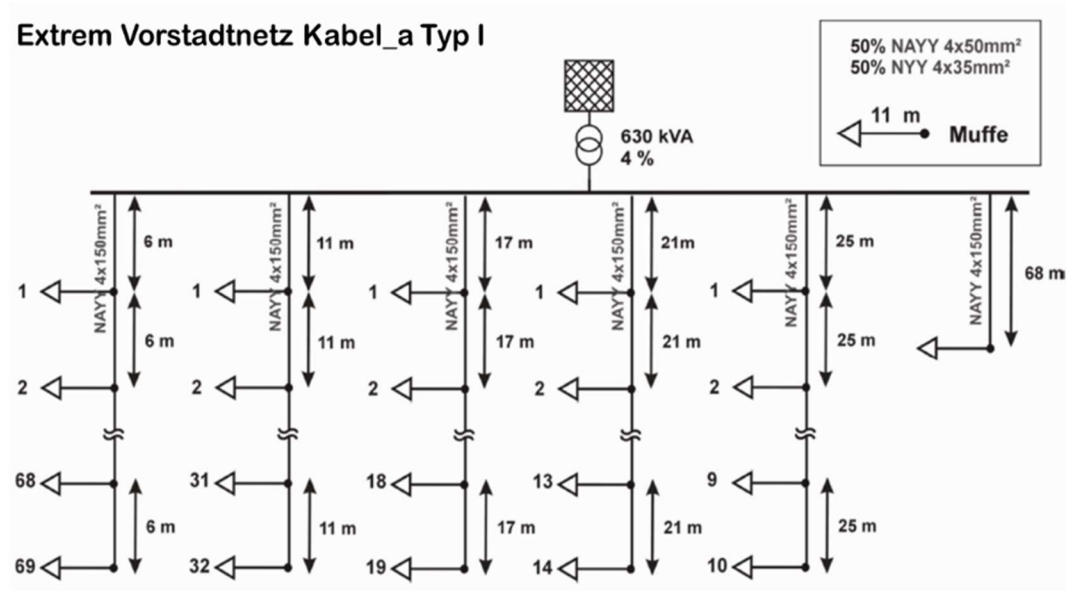


Figure 9-1 Representation of the Kerber network model used in the simulation

There are 145 houses whose loads were replaced by household profile data provided by NEDU (association which represents all the different stakeholders in the energy sector in the Netherlands) for the year of 2022. The normalized profile was scaled up for electricity usage by 4 and 5 person households. The maximum total household load in the network at any given time during the day is about 170 kW. All the households also hosts a roof-top PV installation of rating 3 and 87% efficiency. The generated PV power is based on irradiance data for the city of Utrecht (coordinates: 52.094, 5.040) as taken from PVGIS website. A snapshot of the household and load profiles for four days in February 2020 is shown in Figure 9-2. The wholesale electricity prices used in the simulation are the day ahead price values of electricity for the year 2022.

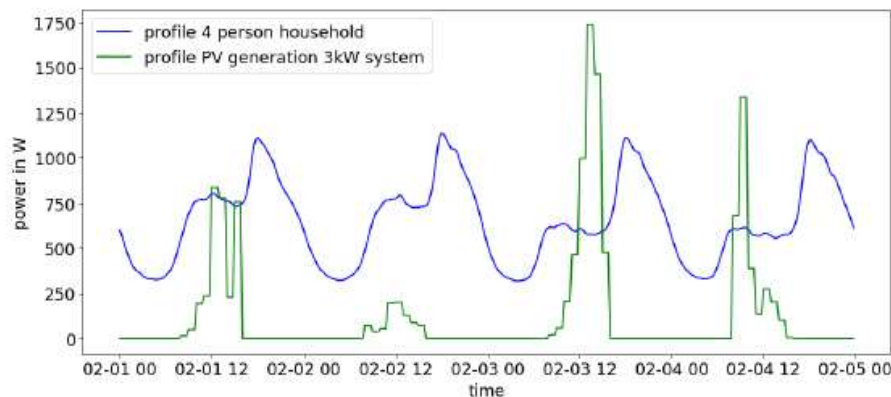


Figure 9-2 Simulated 15 minute resolution power profile of 4 person household load and 3kW PV system.

9.1 Bandwidth model

This example includes EVs in the grid and investigates undesired effects of EV charging on the grid. To fulfil its high energy demands, EVs consume very high amount of power when connected to the grid. A single phase AC charger can have a power rating up to 7 kW while a three phase one can deliver up to 22 kW of electrical power to the EVs. Due to potential high simultaneity of EV charging (and discharging) behaviour can lead to network congestion. This congestion can manifest as voltage problems and overloading of cables and transformers. To reduce congestion, one of the solution is to implement a capacity bandwidth based tariff where customers are contracted to use a specific level of capacity (bandwidth). Within this bandwidth they are free to use the network without any additional charge at any time. However, They have to pay a fine if their load exceeds the capacity they subscribed.

The goal of bandwidth based tariff is to avoid congestion in the network by assets like EVs, which have a very high simultaneity factor. However, in conditions created by high variance in electricity prices, a low enough electricity price compared to the peak price might still create opportunities for EVs break this capacity bandwidth and charge (or discharge) at higher rates overloading the cables and transformers.

For the simulation, a number of EV chargers with a maximum rating of 22 kW are distributed uniformly along each of the feeders of the simulated network. It is important to correctly model the behaviour or charging pattern of the EV, especially the connection start and end times and energy requirements. A simple solution is where the charging start time and energy requirements are either assumed to be same or modelled with a single probability distribution function for all the EVs. Such a model is then unable to capture the charging behaviour of different kinds of users and charging sessions. Another way is to use a portfolio of different charging sessions per EV user. These charging sessions can be categorized based on the details such as charging time (day or night), charging duration (short, medium and long) and hours between sessions. In this simulation model, three types of charging sessions are modelled: Charging during day (uniformly distributed), charging during the work hours (a normal distribution around the office opening and closing hours) and during night (normal distribution for residents to start charging in the evening and depart the next morning). Each charging session is defined by three properties: time of arrival, time of departure and time away (time between two charging sessions). After defining the charging sessions type, different type of EV users are created where each EV user can have a portfolio of different charging session types with varying probability. Since, the idea was to simulate a residential electricity grid, the user types used in the simulation had a higher probability of a charging session during the night.

The charging of the EVs is divided into two phases; constant current and constant voltage phase. During the constant current phase, a constant maximum power is transferred from the chargers to the EVs. In

this work all the EVs are able to charge with 22kW chargers. In the second phase of charging (SOC > 75%), the current and the power transfer decreases exponentially with a form of $I = I_0 e^{-\alpha t}$, where I_0 is the current during constant current phase, t is the SOC and α is the acceptance rate. In this work the acceptance rate after 0.75 SoC is taken to be -0.1. The profile of the EVs for different SoC levels used in this work is shown in Figure 9-3.

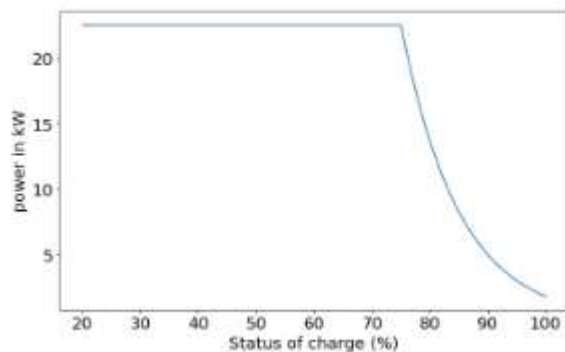


Figure 9-3 Charging profile of EVs

Two types of charging strategy are presented in this report. The base case (EV-basic) has no strategy where the EVs start to charge as soon as they are connected to the charger and charge as fast as possible. The EV depart if they have sufficient energy (given by desired SoC) or at their calculated departure time, whichever comes first. The second charging strategy (EV-smart) is where the EVs after connecting to the charger receive a *perfect* price forecast. It is called perfect because this is taken from the actual price of electricity at the moment. So there are no errors considered in the price forecast. Using the price values during their connection time, the EVs schedule their charging instances such that the cost of charging is minimized.

The results obtained using the above mentioned EV charging strategies are presented by the figures below. No PV panels and house batteries were included in the simulation model for the sake of simplicity. It was also assumed that about 25 % of the household had EVs whose charging behaviour was modelled as discussed above. Figure 9-4 presents the transformer load observed for the basic charging strategy. It can be observed that simultaneous charging of EVs in the evening time causes extreme transformer overloading in the simulated system. For a similar charging behaviour simulation, when smart charging was implemented, this resulted in even higher loading of the transformer. As shown in Figure 9-5, considerable increases in the peak power flows are observed around the local price dips. This was expected to occur due to the simultaneous charging actions activated by the smart chargers based on the low price triggers. Thus a price based smart charging increases the simultaneity factor of the EV charging. There were two constraints for the smart charging algorithm:

- i. the maximum power rating of the EV and EV chargers which was set to be 22 kW.
- ii. The EV has desired level of SoC at the time of departure.

To solve the problem of huge peaks caused by simultaneous charging actions, a capacity bandwidth based tariff was implemented. Each EV user was then subscribed to a penalty-free bandwidth capacity of 8 kW and this was added as one of the constraints. Beyond 8 kW, a penalty of 0.25 euro per kW would be levied on the EV user. This condition was added as a soft constraint in the smart charging algorithm. Such a tariff would limit the severity of the simultaneity problem associated with the smart EV charging based on price signals. However, if the low price values create a situation where there is a financial incentive of breaking the bandwidth (and facing penalty), the EVs (charging algorithm) will take this opportunity and charge together at higher rates. Depending on the number of EVs connected for charging during such time instances, severe loading of the grid may be observed. The results obtained for simulating the same conditions as in the previous cases are presented in Figure 9-6. A reduction in

peaks is observed for most part of the simulation period. However, during local price dips, extreme overloading levels (higher than in case of basic charging) of the transformer is observed.

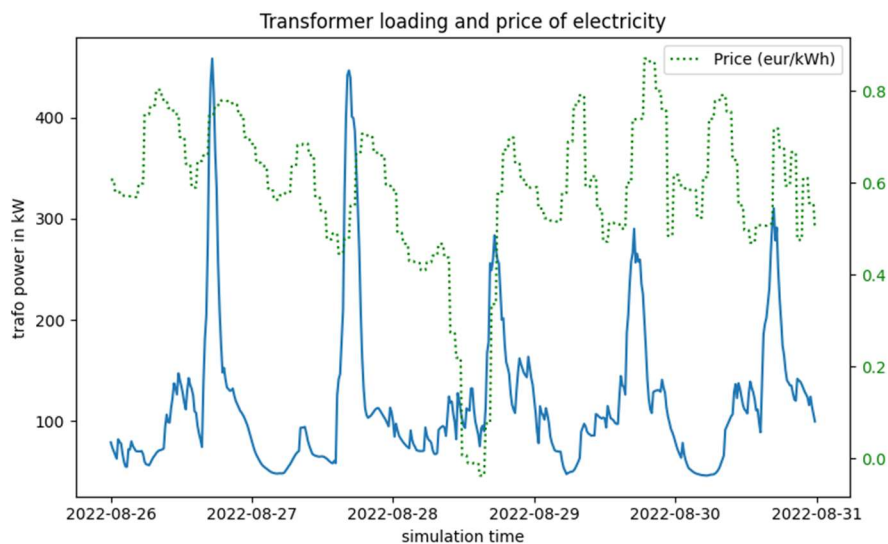


Figure 9-4 Transformer load observed during basic EV charging strategy.

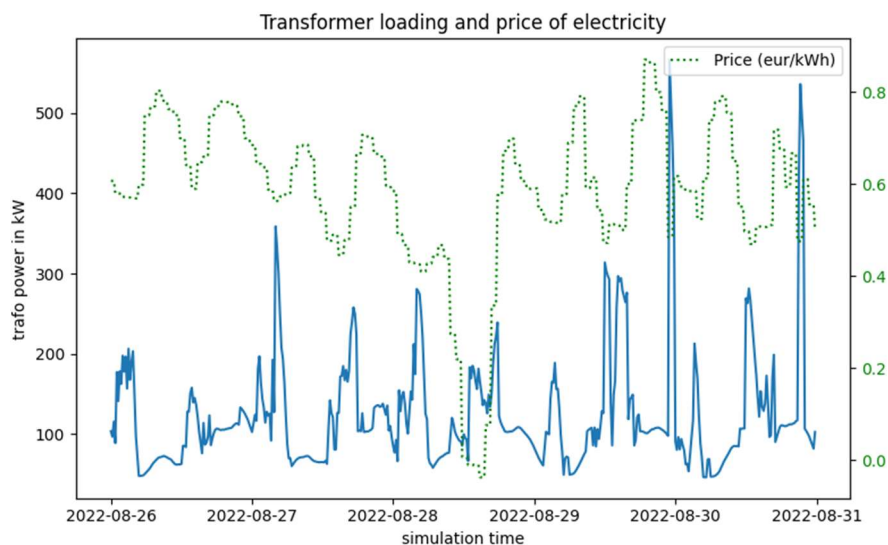


Figure 9-5 Transformer loading in case of smart charging

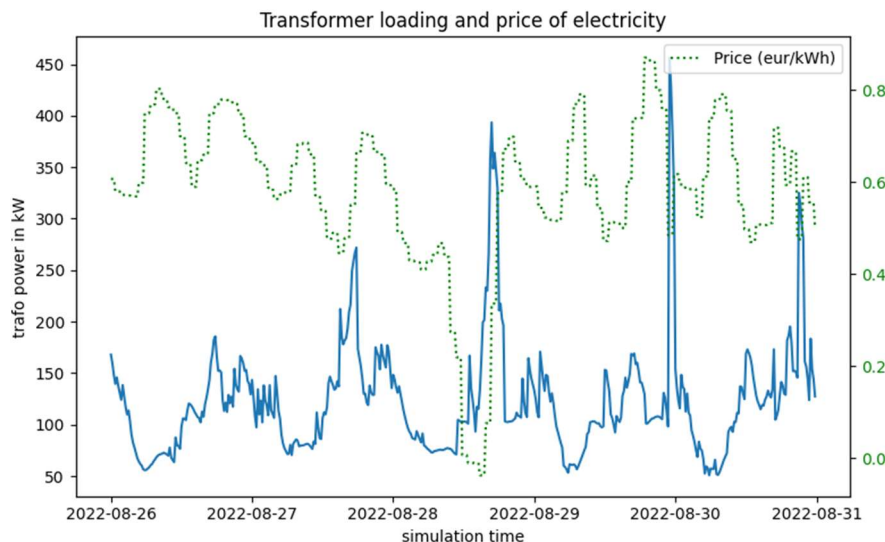


Figure 9-6 Transformer loading in case of bandwidth based tariffs applied with smart charging

This simulation example showed how simultaneity in EV charging along with smart charging algorithms can still pose a challenge for the grid even after implementation of preventive mechanisms like bandwidth capacity based tariffs.

9.2 Statistical assessment of tariff – pricing interaction

The assessment in the previous section demonstrated the possible adverse effects of price incentives for EV charging and the inability of proposed solutions (such as a “bandwidth tariff”) to fully mitigate them. Subsequently, a special-purpose simulation was constructed to investigate this problem statistically across a range of price signals, charging patterns and aggregation levels. The results of this study are summarised here and published as:

Aggregate Peak EV Charging Demand: The Impact of Segmented Network Tariffs

Nanda Kishor Panda, Na Li, Simon H. Tindemans, (accepted for ITEC 2024)

Preprint available: arXiv:2403.12215, <https://arxiv.org/abs/2403.12215>

The simulations use a multi-level segmented tariff (distinguishing low, medium and high load) that – in the absence of other incentives – is generally able to reduce individual and aggregate peak demand. The study investigates the interaction between peak-reducing network tariffs and dynamic energy prices, shedding light on the competition between peak-reducing (network tariffs) and peak-forming (wholesale prices) charging behaviour. Figure 9-7 shows the aggregate charging demand under four different scenarios, based on real charging transactions from over 650 public charging stations (CS) in the Netherlands.

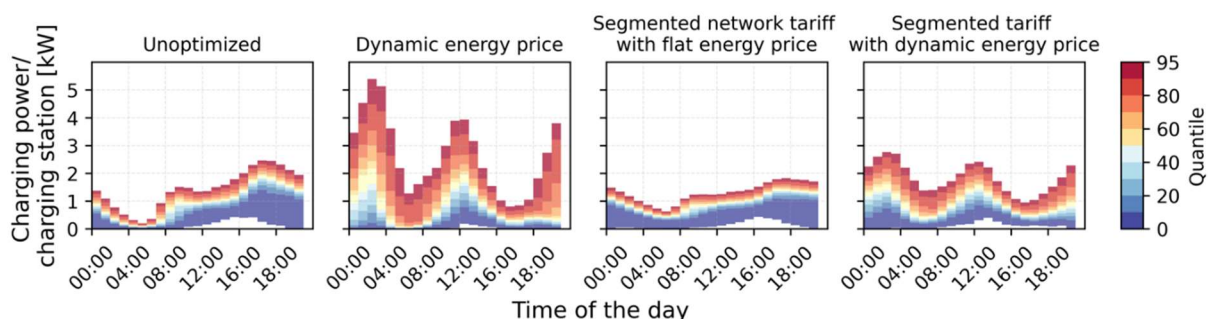


Figure 9-7: Distribution of daily charging demand for different dispatch strategies, using 2022 data.

The results show that the segmented network tariff with flat energy prices results in more diversified load profiles with increasing aggregation compared to cost optimized dispatch based on dynamic day-ahead prices. The average peak demand for charging with segmented network tariffs and flat energy prices is reduced by 49% and 29% compared to optimized charging under dynamic day-ahead energy prices and unoptimized (dumb) charging, respectively.

Focusing on demand peaks, Figure 9-8 summarises the annual peak demand (normalised by the number of charging stations) for different randomly chosen aggregations of charging stations. As expected, in all cases, aggregation increases the diversity of load, reducing the collective capacity needs. Although the combination of dynamic energy prices and the segmented tariff has considerably higher peak demand than both cases without dynamic prices, the reduction relative to *only* dynamic prices is substantial.

Information from these experiments can be used to (1) quantify diversity of EV smart charging schemes, assisting the network operator in planning and scheduling; (2) assess the pros and cons of new capacity-based tariff schemes.

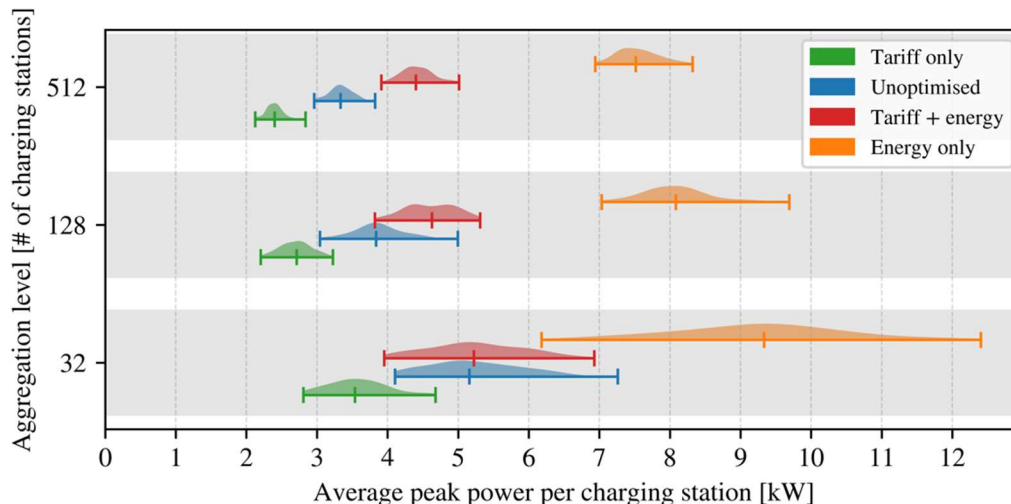


Figure 9-8: Distribution of peak charging power per charging station for different aggregation levels.

9.3 Home batteries

This example includes household batteries and investigates any undesired effect on to the grid caused by them. Batteries were included in 50% of the households. The house battery system has two goals. Primarily, for an end-user, maximize the self-consumption of energy from the rooftop PV panels. Secondly, minimize the electricity bill. From the grid's perspective, such user level objectives, should lead to benefits like, reduction in voltage rise during peak sunny hours, local use of generated electricity and reduction in congestion. To achieve its objectives, input information for the battery management system are: expected PV generation, expected household load, electricity retail price and state of charge of the battery. Using these input signals the battery is scheduled to charge or discharge at particular time instances. These charging and discharging decisions were constrained by the following conditions:

- i. The energy stored in the battery may never exceed the capacity of the battery
- ii. The energy stored in the battery may never be less than 5% of the capacity of the battery
- iii. The energy stored in the battery at a given time instant (t_n) is given by the sum of energy stored at time (t_{n-1}), energy charged from the PV system, energy charged from the grid, energy discharged to the grid and energy discharged to the household load. The charging and discharging efficiency is 90%.
- iv. The maximum rate of (dis)charging is limited by the rating of the converter.
- v. The maximum rate of charging is limited by the rating of the converter.
- vi. The PV energy transferred to the household load or the battery is limited by the power rating of the PV system.
- vii. The household load is always supplied either by the battery, PV system or by the grid.

Multiple scenarios were simulated to see the interaction between the control operation of the battery management system and the other system variables like electricity price, PV system size, battery capacity, time of the year etc. Particularly interesting to explore was the expected synchronous behaviour of the battery system in response to the price signals during summer and winter months. In summer the battery system is easily charged due to available solar energy via the PV system. In case of high power prices, these batteries can discharge synchronously causing sudden and large power-flow

reversal. This can lead to reversal of power at the distribution transformer feeding the simulated grid. In winter however, there might be insufficient solar power to charge the house batteries. This might result in batteries operating at lower state of charge. In such conditions, if the electricity price reach certain lower levels, all the batteries might take this opportunity to charge synchronously causing a surge in the load of the grid.

Different parameters used in the simulation model are listed in the Table:

Battery Capacity	10, 15 and 20 kWh
Battery Charging/Discharging rate	50% of the capacity charged/discharged in 1 hour
PV system	3 kW
Tax on electricity purchase	0.15 eur/kWh
Optimization period	1 day
Percentage houses with PV system	100%
Percentage houses with battery	50%

Battery capacity should be enough to enough to store the PV energy during the summer months and then supply the household. However higher capacity of batteries might be unused during the darker winter months. This would result in under-utilization of the battery and a waste of capital cost. For example, the status of charge (SOC) plots of three different battery capacity used in the simulation are shown by Figure 9-9 to Figure 9-11 for a week during the summer. The first day of the simulation results have not been included to remove the effect of model initialization. Apart from the SOC values, the non-flexible household load, generated PV power and the wholesale electricity price values are also plotted. The electricity sales tax on top of the wholesale price is 0.15 eur/kWh. It can be observed that the 10 kWh battery get full and loses opportunity to charge the generated energy from the 3kW PV system. The 15 kWh battery is of decent size to capture the PV energy. The 20 kWh battery is never charged near fuller capacity and is too big for the system of PV and the household loads. The battery charging and discharging rates are often dependent on the capacity of the battery

The results achieved using a battery of capacity 15 kWh in summer and winter months is presented and discussed further. Sunny days during the summer-time and days without sun during the winter-time are common. However, to look for the boundary conditions, the simulation periods had to be carefully selected to get the required conditions.



Figure 9-9 SOC of the 10 kWh house battery during a week in summer

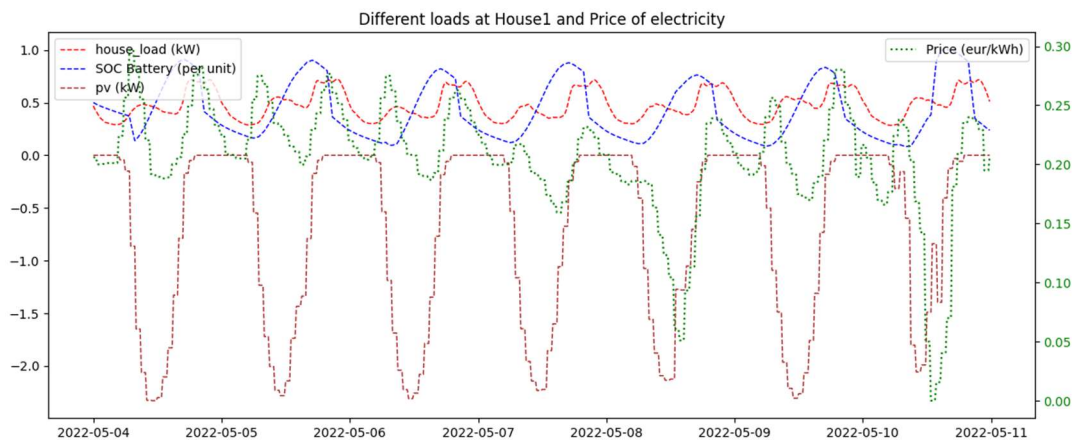


Figure 9-10 SOC of the 15 kWh house battery during a week in summer

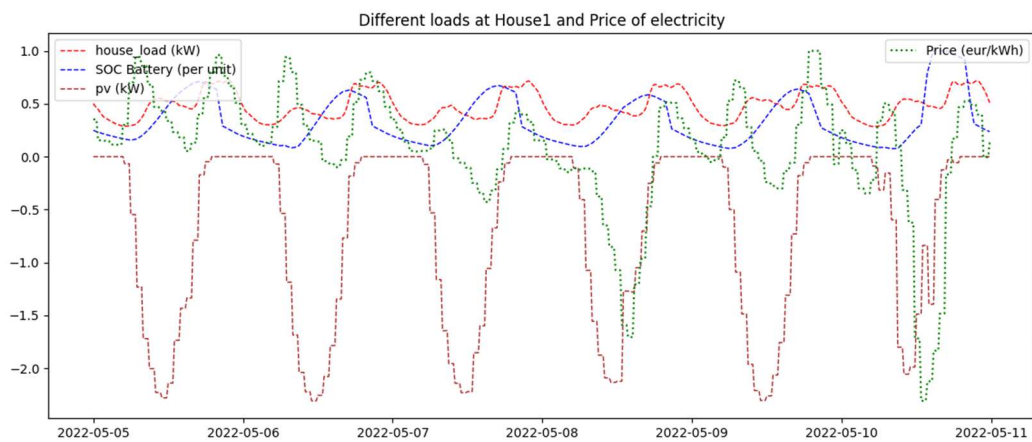


Figure 9-11 SOC of the 20 kWh house battery during a week in summer



Figure 9-12 Transformer loading values in case no batteries are included in simulation and 100% of households have PV generation.

Figure 9-12 shows transformer loading values in case no batteries are included in simulation and 100% of households have PV generation. It can be observed that due to PV generation, not only reverse power flows in the transformer, but it is also loaded over its capacity on every day when the sun shines.

Figure 9-13 to Figure 9-16 presents the results for the simulation period in winter. All the loads (positive power) and generation (negative power) at bus 1-69 for the period of simulation during the winter are plotted in Figure 9-13. Bus 1-69 is where the house 1-69 connects to the grid. From Figure 9-14, it is observed that, in the winter the battery is not cycled (charging and discharging) due to the lack input from the PV system. The SOC of the battery thus remains on the lower end. This, however, leaves a possibility for battery charging if there is enough variation in the electricity process are expected. The battery can then charge during the low price periods. This sort of sudden peak in battery charging load is observed towards the end of the simulation period in Figure 9-13.

If there are multiple batteries in the grid with similar goals to maximize profits or minimize electricity costs, they will impose similar load on the grid triggered by the price values. Such individual but synchronous actions by the house batteries can be create conditions like overloading of transformers and cables or reversal of power flow. Conditions like overloading and reversal of power flow may then result into voltage dips or swells in the network and might trigger the protection devices cutting the local or regional grid from the rest of the network. As observed in Figure 9-15 and Figure 9-16, the simultaneous charging action of the batteries results in sudden increase in the transformer loading and a voltage dip for the duration of charging action. These peaks are significantly higher than the peaks observed when no batteries were included in the simulation.

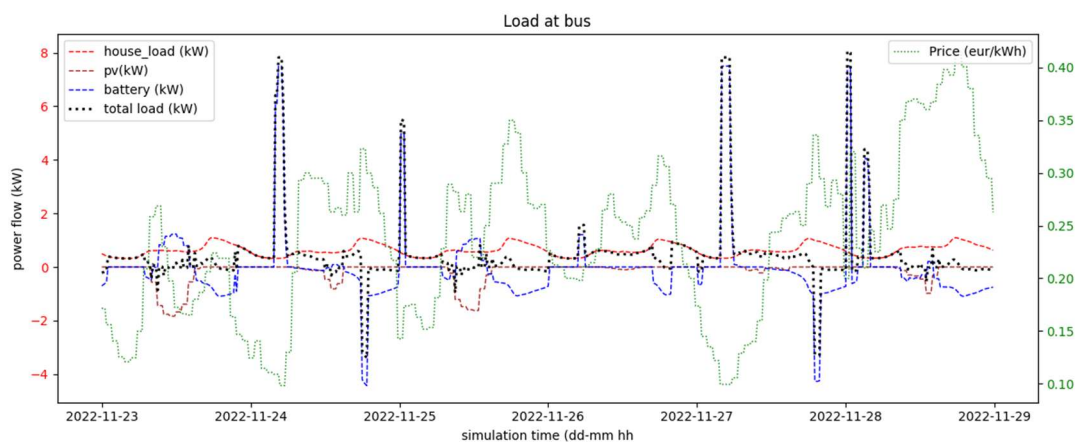


Figure 9-13 Different loads of a household (number 1-69) during a week in winter

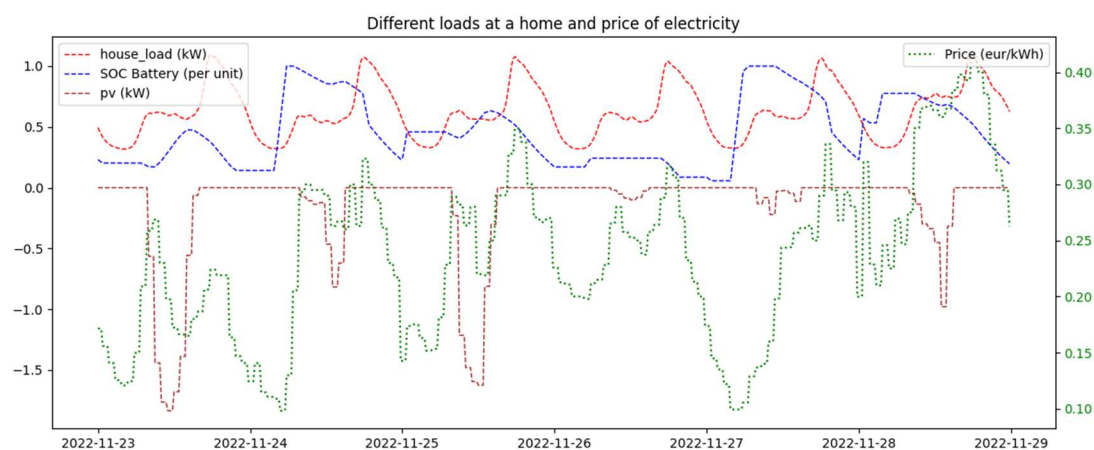


Figure 9-14 SOC of the 15 kWh house battery during a week in winter

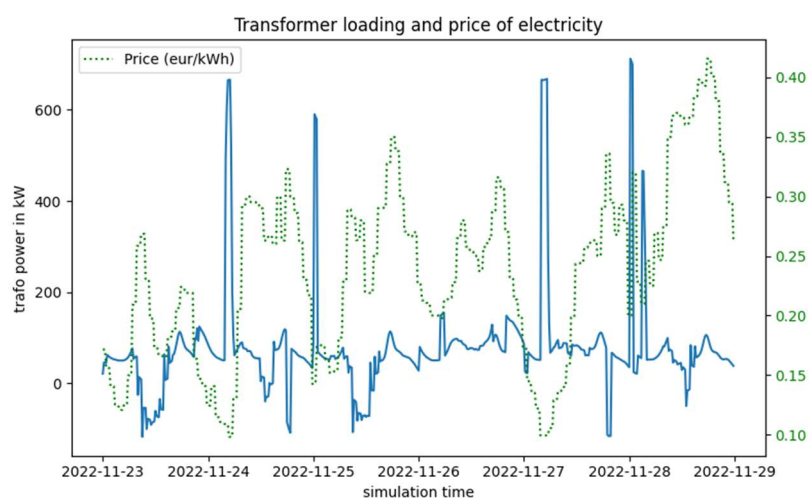


Figure 9-15 Distribution transformer loading during a week in winter

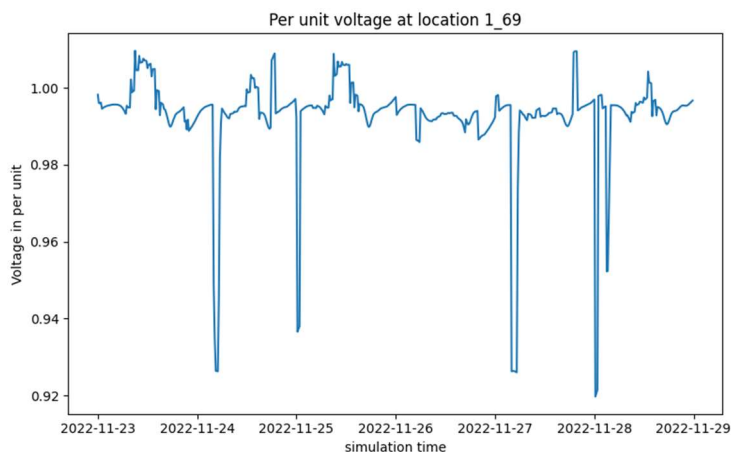


Figure 9-16 The voltage at the connection point of house 1-69 during a week in winter.

Figure 9-17 to Figure 9-19 show the simulation results for the same system during a week in summer. In the summer when the sun is abundant, the charge cycle of the battery is frequent (almost daily). The battery charges regularly during the day storing the generated PV power which is free of charge. This energy is then discharged at the time with highest electricity prices. The instances of discharging power can be observed in the Figure 9-17. During discharging, the battery is an energy source from the point of view of the grid. Synchronous generation activity triggered by the price signals lead to sudden reversal of power flow and a swell in the voltage in the grid. These phenomena are to be seen in Figure 9-18 and Figure 9-19.

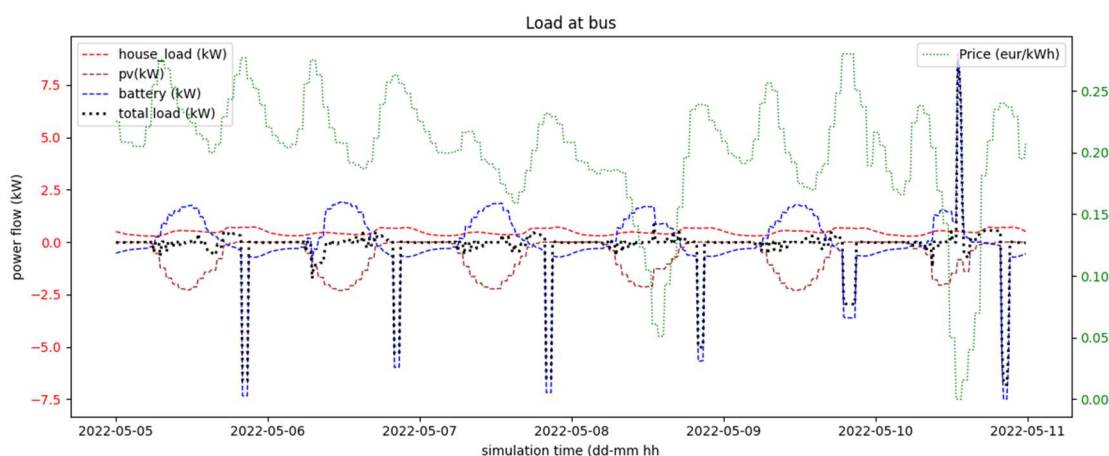


Figure 9-17 Different loads of a household (number 1-69) during a week in summer.

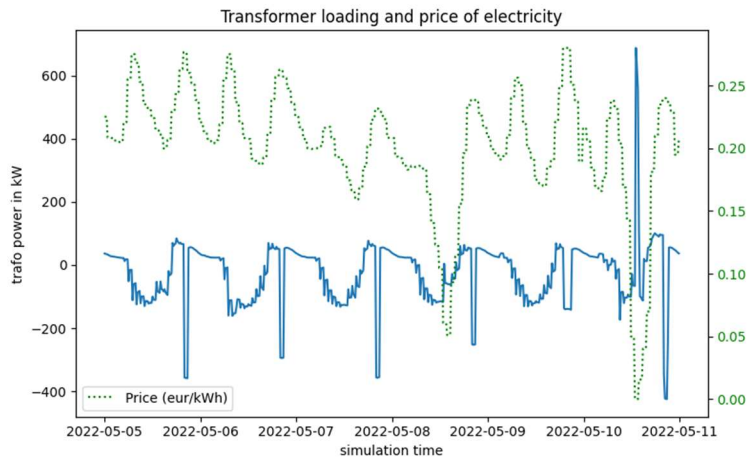


Figure 9-18 Distribution transformer loading during a week in summer.

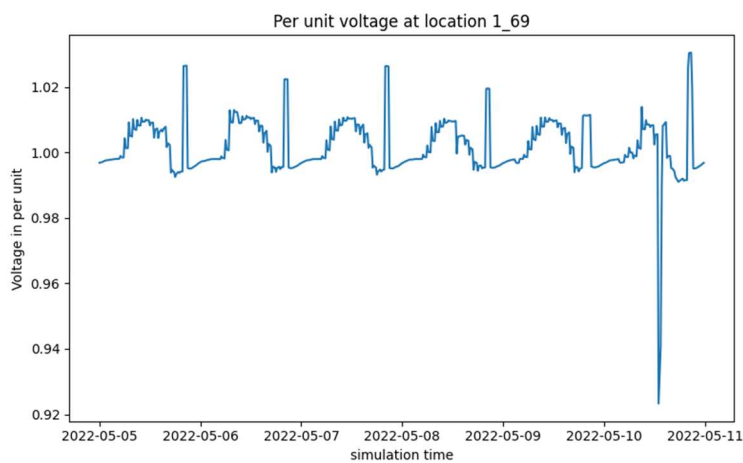


Figure 9-19 The voltage at the connection point of house 1-69 during a week in summer.

This simulation example presented the effect of simultaneity in house batteries. The goal of the house battery was to catch the generated PV power and use it later to supply the household loads. From the grid's perspective, this would eliminate the problem of huge reverse power flow. However, triggered by price values, the simultaneous charging in winter and simultaneous discharging in summer even though for shorter periods, resulted in more severe overloading and power reversal than when no batteries were used.

9.4 Increase-decrease gaming for redispatch markets

Redispatch markets are an important tool for network operators to mitigate (projected) congestion resulting from the day-ahead market clearing. Because the network operator is obliged to resolve congestion, this results in an opportunity for market actors to extract profits, resulting in so-called increase-decrease gaming. This has been observed both in theory and practice for transmission systems.

The following describes a study that is aimed at identifying the risk of increase-decrease gaming for electricity distribution networks with large numbers of participating users. The main aim was to identify inherent risks of gaming *without* explicit price manipulation, i.e. to see whether *involuntary* gaming can result from collective actions. This was studied in an analytical manner using mean-field game theory. It summarises the results from the following papers:

- "Mean Field Game for Strategic Bidding of Energy Consumers in Congested Distribution Networks", A Silani, SH Tindemans, IEEE CDC 2023, doi: 10.1109/CDC49753.2023.10384016.
- "Stochastic Mean Field Game for Strategic Bidding of Consumers in Congested Distribution Networks", A Silani, SH Tindemans (submitted). Preprint available: <https://arxiv.org/abs/2403.11836>
- "Mean-Field Stackelberg Game for Mitigating the Strategic Bidding of Energy Consumers in Congested Distribution Networks", A Silani, SH Tindemans (submitted)

The problem is modelled as shown in Figure 9-20. It consists of a day-ahead market (first stage) and redispatch market (second stage). Energy consumers first purchase energy on the day-ahead market, according to their utility and market expectations. If the DSO detects congestion after also taking into account inflexible demand, it requests redispatch bids. This model has a multi-level (leader-follower) structure of a reverse stochastic Stackelberg game. Under the simplified conditions of the model, it can be solved analytically to show that there is unique Nash equilibrium: a strategy that all agents will converge to if they act strategically and with full information.

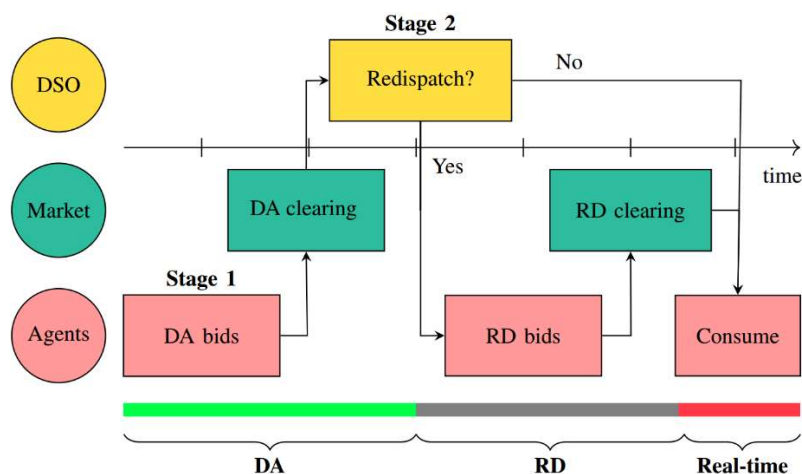


Figure 9-20: Market model.

Based on the analytical results (proof of existence of an equilibrium and methods to compute it), the behaviour of strategic consumers and their impact on the redispatch costs was analysed for a number of cases. It was compared to scenarios where consumers do not anticipate the redispatch market (naive scenario).

Figure 9-21 illustrates the impact of collective strategic trading on the welfare of individual agents. For the fixed day-ahead price (ii), a small proportion of agents – those with utilities just below the cost of day-ahead energy – benefit from anticipating the redispatch market. Figure 9-22 shows that this comes

at the expense of large costs for the DSO (*RD revenue* for the agents). The average welfare of all agents is improved, at the cost of the DSO.

The case with demand-dependent day-ahead prices (i) shows a different picture. The strategic behaviour of agents that benefit from the redispatch market (red curve above the blue curve) results in a higher day-ahead clearing price, which *reduces* the welfare of agents who assign a higher inherent value to the use of electricity (red curve below the blue curve). In this case, Figure 9-22 shows that the average welfare of the agents has *decreased* compared to the case without strategic behaviour – and this is in addition to the costs incurred by the DSO operating a redispatch market.

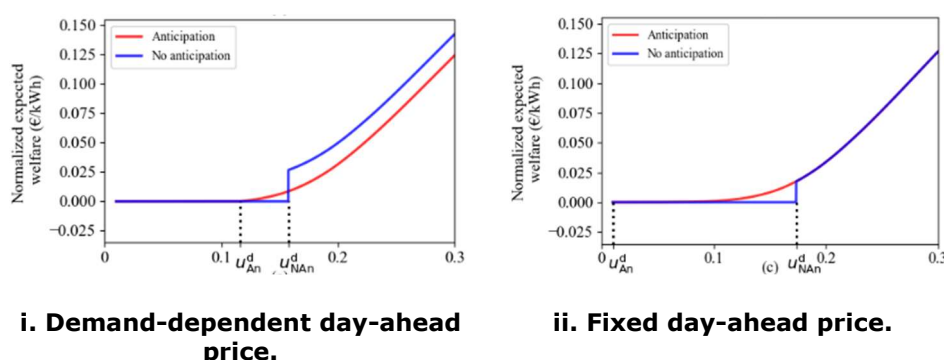


Figure 9-21: Expected welfare of agents as a function of their inherent utility (horizontal axis). Two cases are shown, with demand-dependent and fixed day-ahead market prices.

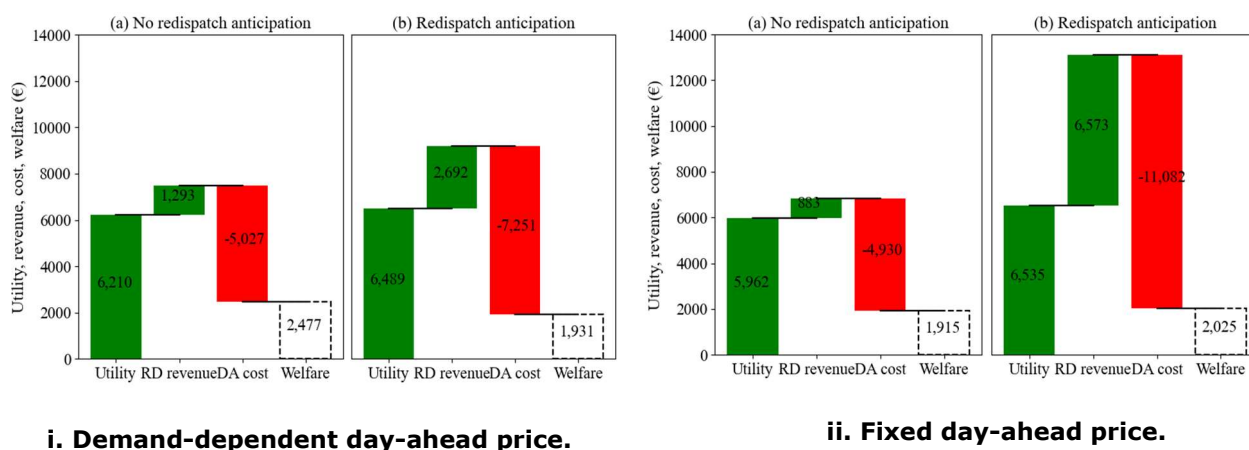


Figure 9-22: The total expected welfare, utility, redispatch revenue and day-ahead cost with medium network capacity and stochastic inflexible load.

An extension of the model was used to investigate a possible mitigation mechanism. In this case, the DSO (or another entity) is able to make the first move by announcing a price premium for the day-ahead market, thus discouraging excess energy purchases. The price is chosen such that a balance is struck between the social welfare of the agents and the costs incurred by the DSO. Figure 9-23 shows that a new balance can indeed be found. It reduces agent welfare (though mostly by reduction of profits extracted from the DSO) and significantly reduces DSO expenditure.

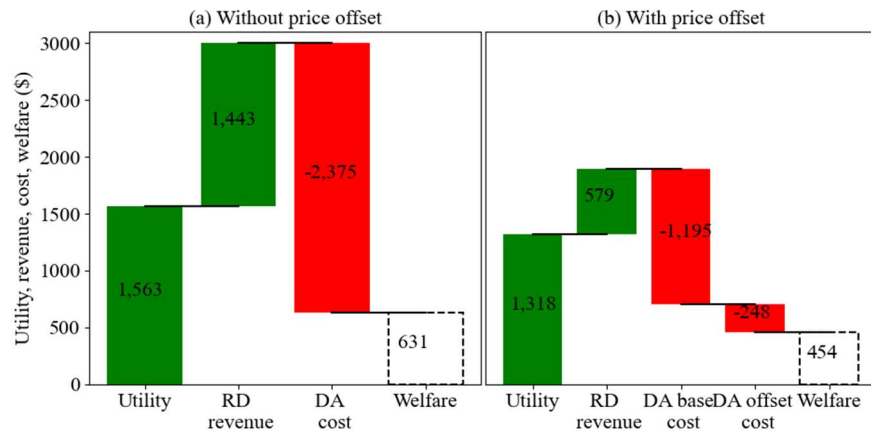


Figure 9-23, Total consumers' expected welfare, utility, redispatch revenue and day-ahead cost: (a) price offset is not considered and (b) price offset is considered.

10 APPENDIX: LIST OF SYSTEM LOSSES IN THE POWER SYSTEM

10.1 Stakeholder analysis (ref to ENTSO-E)

To identify losses on power system level, the roles and purpose of all relevant stakeholders in the power system should be considered.

General:

- Loss of life/ significant injury
- Loss of property/ damages

For each of the stakeholders

- Loss of mission/purpose
- Financial losses
 - Unexpected or /unsustainable high costs
 - Unexpected or unsustainable low income
- Loss of reputation
 - Due to other losses
 - Legal

While the general losses can be considered of a higher priority than the stakeholder losses. In a multi-stakeholder system such as the power system, conflicts between potential losses might occur (and are in place). Circuit breakers for example are meant to disrupt the power flow in order to prevent further damages to equipment or to prevent hazardous situations that could lead to significant injuries or loss of live.

10.1.1 List of most relevant stakeholders and their main possible losses

Stakeholder	Role	Main interest (priority might differ per stakeholder)
End-user	Use power for an (acceptable) price	- Use electricity - Have a low bill - Have a predictable bill (risk) - Sourcing (have electricity from a specific (green) source)
Producer	Sell power for an (acceptable) price	- Predictable income - Profit/margin - Predictable costs -
Transmission system operator (TSO)	Responsible for expanding and maintaining the distribution network Responsible for system operations (a.o. the balance withing an imbalance settlement period (ISP))	- Maintain system integrity - no power outage - stability - system operation - keep load within the constraints of the network
Distribution system operator (DSO)	Responsible for expanding and maintaining the distribution network	- Distribution system integrity - Keep load within the constraints of the network -

Balance responsible party (BRP)	Financial responsible for balancing feed-in and offtake of their portfolio per imbalance settlement period (ISP)	- Have low imbalance risk -
Energy Supplier	Procure energy for their customers and invoice them for all costs related to their electricity bill.	Have competitive bill
Balance Service Providers (BSP)	Offers balancing energy and/or capacity to the TSO	
Congestion Service Provider (CSP)	Offers congestion management services to the TSO or DSO	
Aggregator	Bundles flexibility from multiple small consumers and producers to be offered on the wholesale, balancing or congestion market, or directly to market parties (BRP, or BSP or CSP)	
Regulator	Ensure an efficient power system	
Metering company	Maintain the required metering infrastructure and collect the required metering data necessary to facilitate the market processes and remuneration between stakeholders.	
Other		
- Energy service company		
- Aggregator		
- IT solution supplier		

Most stakeholders in the above table can be split into two: the directly involved stakeholder and by standing stakeholders. However, this is mostly relevant for the non-regulated stakeholders (i.e. end-users, producers and other market players).

For example there are end-users using an algorithm to automatically charge their electric car, and his neighbour that might get affected by this because of power quality issues that can emerge among others because of this charging.

Goals/possible losses:

- Customer minutes lost
 - Over voltage
 - Over current
- Cost
- Power quality (dips, fluctuations, harmonics etc.)

11 APPENDIX: LIST OF KNOWN INTERACTIONS THAN CAN LEAD TO UNDESIRED EFFECTS

11.1 List of possible interactions

Until now there have been identified five categories of possible interactions control and optimization algorithms in electricity systems. These categories should be expanded upon if further experience is gained:

1. *Triggers for simultaneous behaviour*,
For example triggered or synchronized by an event or signal (such as a uniform price).
Examples are Lighting strikes, price signals, outages (e.g. cold pick-up, such as heat pumps all picking up together to catch up with the heat demand after a power outage, or the synchronisation of the legionella protection).
Special care should be taken when there is accumulation/storage and a chance for simultaneous release of energy.
2. *Second and higher order system control instabilities*
These are often already the study of power electronics, for example for grid support such as inertia, frequency support and black start capabilities from variable renewable electricity sources, such as (grid forming) wind farms and can be examined for example using Bode and/or Nyquist diagrams,
3. *Undesired triggering or undesired failure of triggering of controls or protection* (e.g. anti-islanding, fault ride through).
An example of such an undesired mechanism is the failure of the anti-islanding protection of PV systems because they are fed by their neighbouring PV installations.
4. *Harmonic interferences*
For example a possible interference between sampling of measurements and harmonics
5. *Incentives counteracting each other without limitations/ increase-decrease gaming* (e.g. with reinforcement learning)

12 APPENDIX: DESCRIPTION OF THE RESULT IN THE GO-E PROJECT PLAN

Result 3.3: A "recommended practice" for determining the possible negative side effects of a large-scale implementation of new optimization or control algorithms.

This pre-standard includes a categorization of possible failure mechanisms due to interference from different (types of) algorithms to unlock flexibility in large-scale implementation. In addition, it includes the methods and processes how to validate these risks per algorithm. The recommended practice includes at least:

- A categorization of mechanisms and algorithms based on the interaction and data exchange.
- A list of modelling and simulation-based interference mechanisms that underlie the negative effects of automated algorithms, including criteria and conditions when they may manifest.

This result is achieved through the following activities:

Activity 3.4.1

*Participating parties: **DNV GL**, TNO, TU Delft, TU Eindhoven*

Inventory and categorization of the mechanisms and / or algorithms to be validated on the basis of their interaction with the environment. This builds on the mechanisms and algorithms for the DSO from activity 3.2, and algorithms for unlocking flex by other parties, including from WP1 and WP2. The algorithms are typified on the basis of the interaction and data exchange with the system (including other algorithms and mechanisms).

Activity 3.4.2

*Participating parties: **TU Delft**, DNV GL, TNO, TU Eindhoven*

Modelling and categorization of the identified interference mechanisms, including deliberate interference (gaming). For each mechanism, an estimate is made of the probability of occurrence and the consequences. Identified problems of interaction between algorithms and with the system are analysed and categorized according to interference and failure mechanism by means of modelling the operational and critical reference environment of these algorithms. This is done in conjunction with the modelling activities in activity 3.3. Negative effects of interactions are validated through modelling and simulations, and criteria and typical circumstances are drawn up in which they can occur.

Activity 3.4.3

*Participating parties: **DNV GL**, TNO, TU Delft, TU Eindhoven*

Generalization of the used process and methods to a "recommended practice" (pre-standard) for validating optimization and control algorithms and their impact on the electricity system.

Above result and progress can be verified through sources of information:

- List with inventory and characterization of the mechanisms and / or algorithms to be validated. (Result activity 3.4.1, M12, lead DNV GL)
- List of potential risks and (critical) reference conditions and associated models. (Result activity 3.4.2, M30, lead DNV GL)
- Table with type of algorithms (on x and y axis) versus interference and failure mechanisms (on z axis) (Result activity 3.4.2, M36, lead DNV GL)
- Results of actual simulations and validation in the labs of DNV GL, TUE and TUD. (Result activity 3.4.2, M36, lead TUD)
- Recommended practice (Result activity 3.4.2, M36, lead DNVGL)
- Various scientific papers (M36, lead TUD, TUE)

Depending on the risks of interference from control algorithms in the electricity system, which are apparent from these results, the recommended practice can develop into a standard that can be used by independent parties to map the risks of new algorithms in the event of large-scale implementation, or by third parties creating algorithms that do not compromise the stability and integrity of the electricity system. Some examples of unwanted interferences that fall within the scope are:

- Instability as a result of simultaneous behaviour, synchronized by, for example, price incentives.
- Interference between control systems leading to oscillations (2nd order instability).
- False claims about the effectiveness of optimization algorithms that are not fulfilled in practice because they are nullified by other mechanisms / algorithms.
- (Learned) undesirable behaviour of reinforcement learning systems (which may exploit mechanisms of other parties, such as DSO's improper learning).
- Mechanisms and algorithms that counteract each other with undesirable effects.